PROTEINS

STRUCTURE , CLASSIFICATION AND FUNCTION

Learning Objectives

- Define Proteins
- Classify proteins on the basis of structure , shape, nutrition, function etc.
- Enlist the function of different proteins such as hormones, enzymes, cytoskeleton, cell membrane proteins, antibodies etc
- Discuss structure-Function relationship with examples

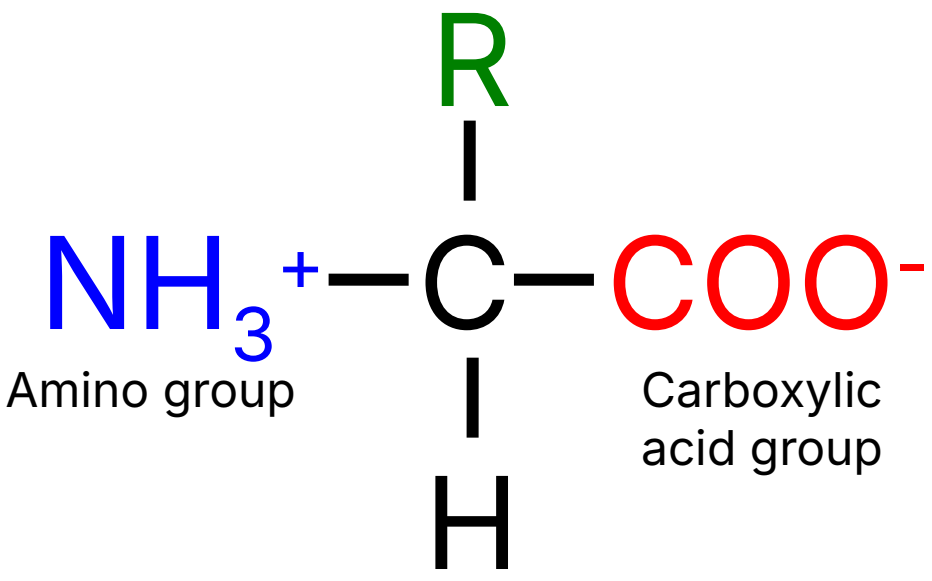
Proteins play key roles in a living system….

If there is a job to be done in the molecular world of our cells, usually that job is done by a protein.

Examples of proteins include

- Hormones acting as messengers;
- Enzymes speeding up reactions;
- Cell receptors acting as ‘antennae’;
- Antibodies fighting foreign invaders;
- Membrane channels allowing specific molecules to enter or leave a cell;
- They make up the muscles for moving;
- Let you grow hair, ligaments and fingernails;
- Let you see (the lens of your eye is pure crystalized protein).

Amino acid: Basic unit of protein

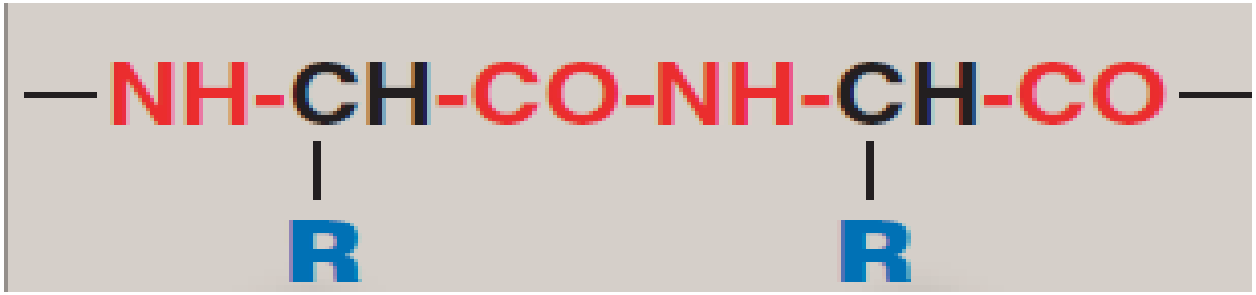


An amino acid

Different side chains, R, determin the properties of 20 amino acids.

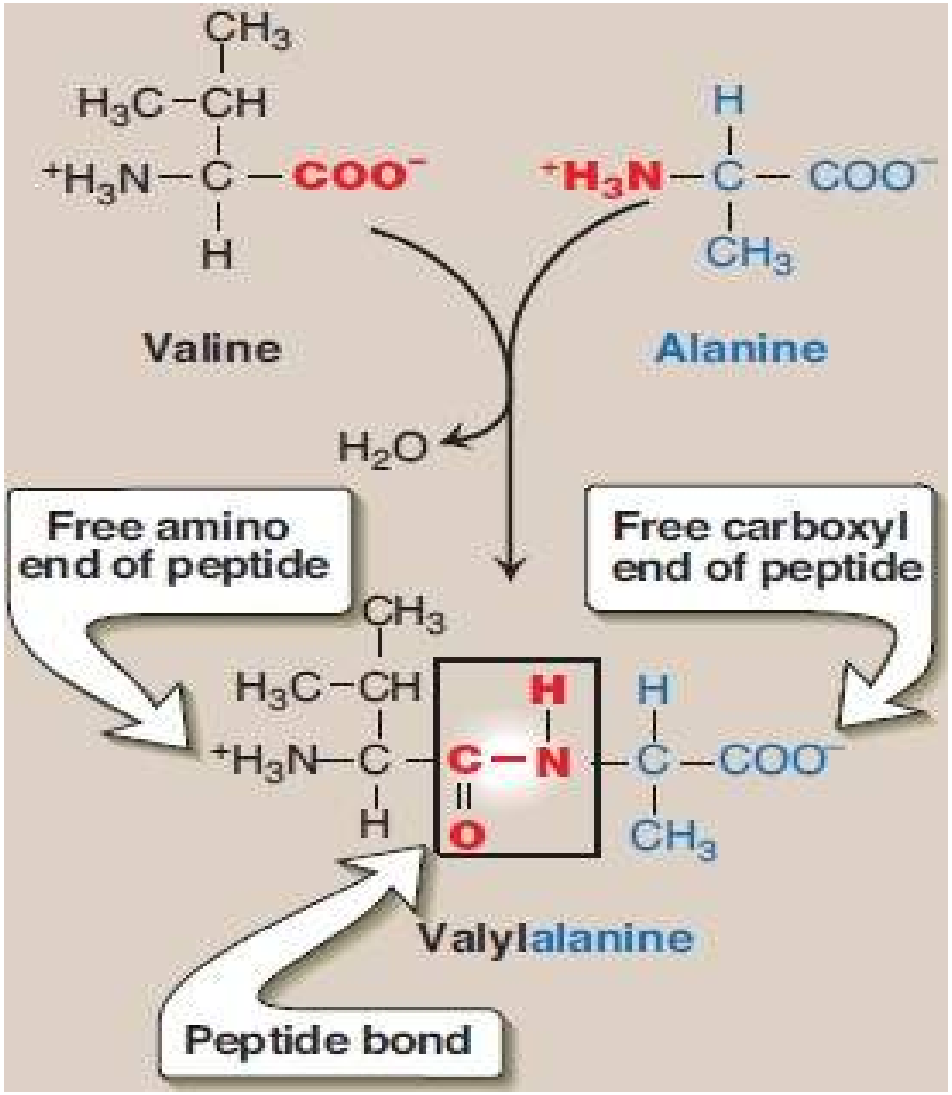
Proteins

- Proteins are the polymer of amino acids
- The 20 amino acids commonly found in proteins are joined together by peptide bonds
- Polypeptide backbone is the repeating sequence of the N-C-C-N-C-C… in the peptide bond



The peptide bond

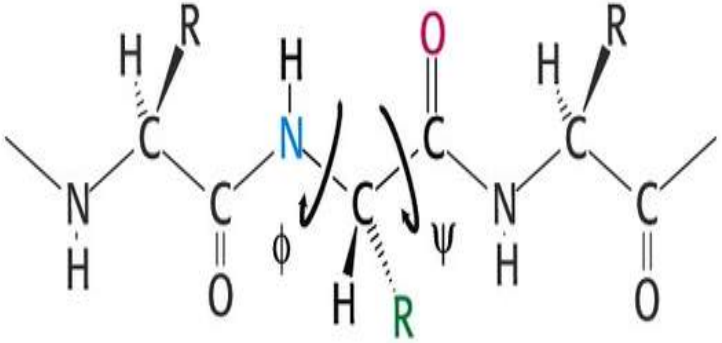
- Is a **covalent** bond
- are amide linkages between the α-carboxyl group of one amino acid and the α-amino group of another
- Peptide bonds are not broken by conditions that denature proteins



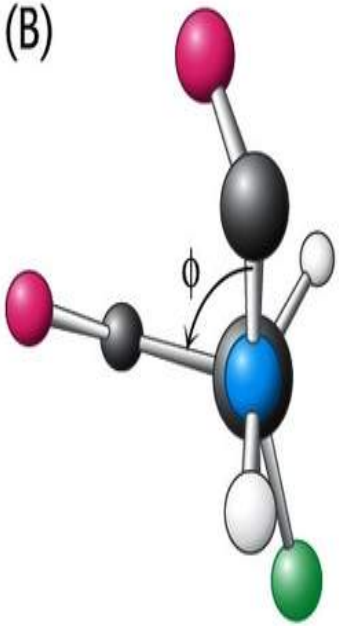
Rotational Properties of Peptide Bonds

- Rotation around the peptide bond is not permitted. It is rigid
- Rotation around bonds connected to the alpha carbon is permitted
- ϕ (phi): angle around the α-carbon—amide nitrogen bond
- ψ (psi): angle around the α-carbon—carbonyl carbon bond
- In a fully extended polypeptide, both ψ and ϕ are 180°
- Allows the polypeptide chain to assume a variety of possible configurations.

(A)

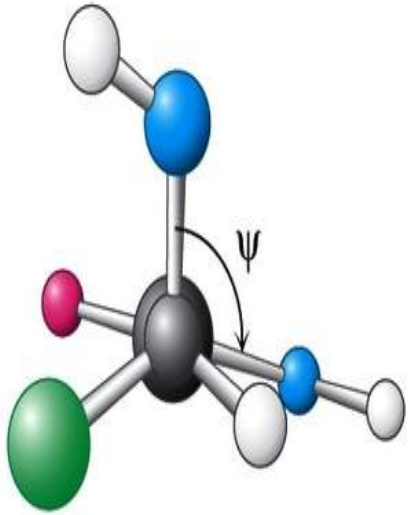


(B)



$\phi = -80°$

(C)



$\psi = +85°$

CLASSIFICATION OF PROTEINS

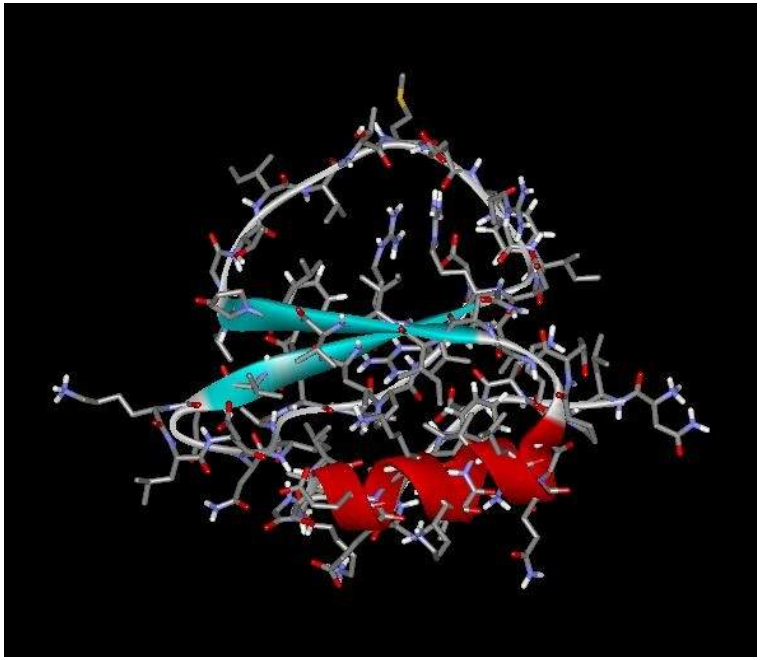
- Structure
- Physico-chemical aspects (Simple, Compound, Derived)
- Nutrition (complete and incomplete)
- Shape (Fibrous , Globular)

Each Protein has a unique structure

Amino acid sequence

NLKTEWPELVGKSV
EEAKKVILQDKPEAQI
IVLPVGTIVTMEYRID
RVRLFVDKLDNIAEV

Folding!



Why Investigate Protein Structure?

Proteins are complex molecules whose structure can be discussed in terms of:

primary structure
secondary structure
tertiary structure
quaternary structure

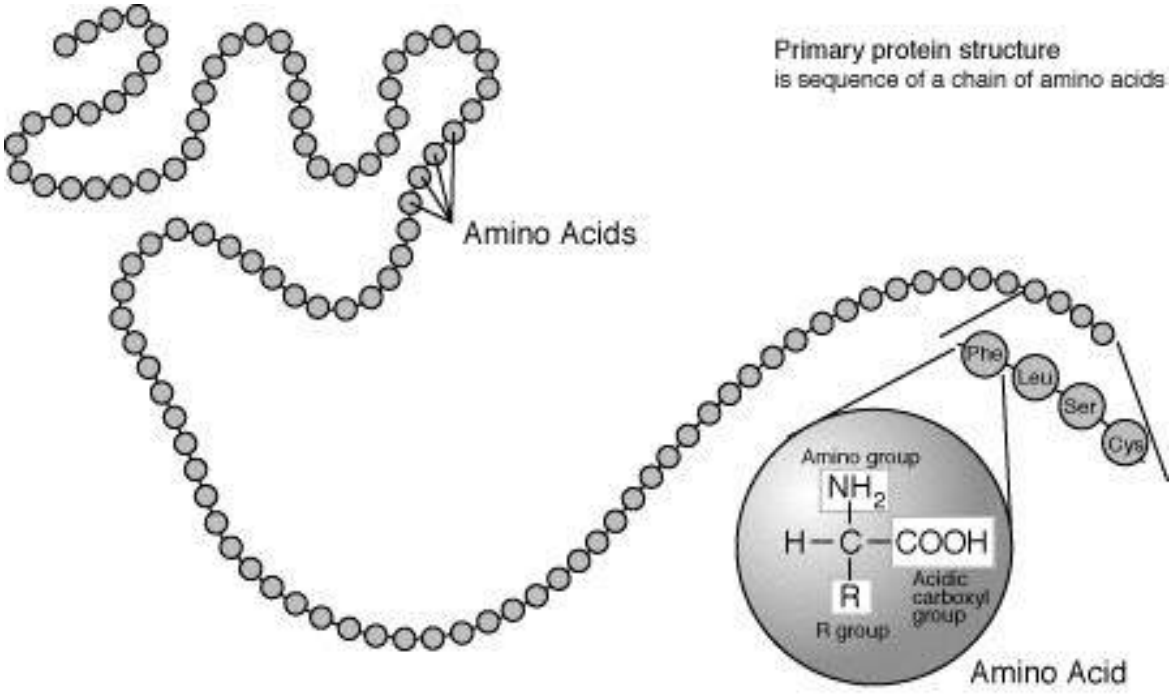
The structure of proteins is important as the shape of a protein allows it to perform its particular role or function

Structural Classification

- Primary
- Secondary
- Tertiary
- Quarternary

Protein Primary Structure

The **primary structure** is the sequence of amino acids that are linked together. The linear structure is called a polypeptide



http://www.mywiseowl.com/articles/Image:Protein-primary-structure.png

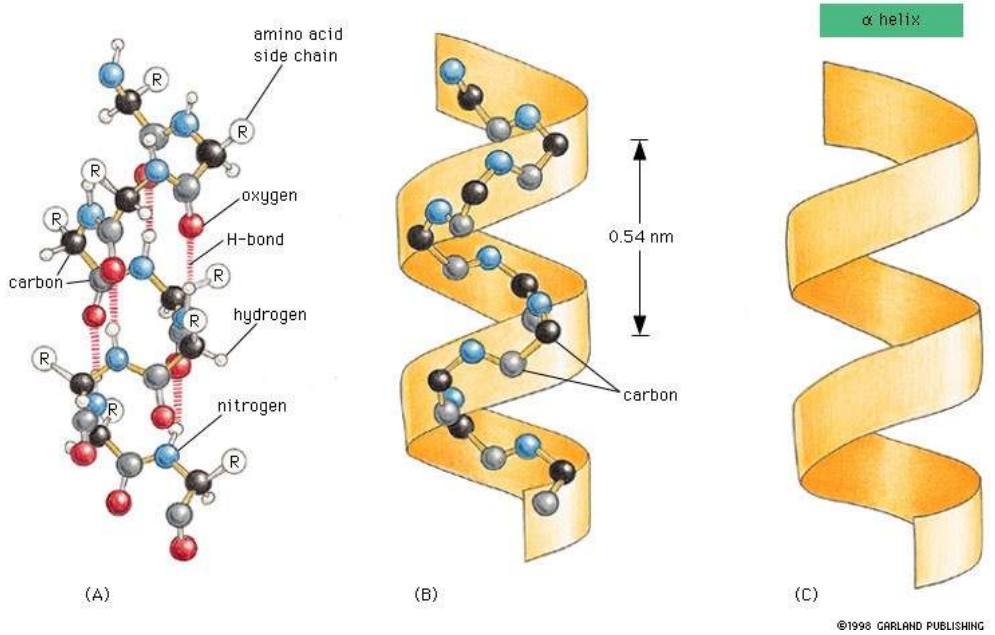
PROTEIN FOLDING

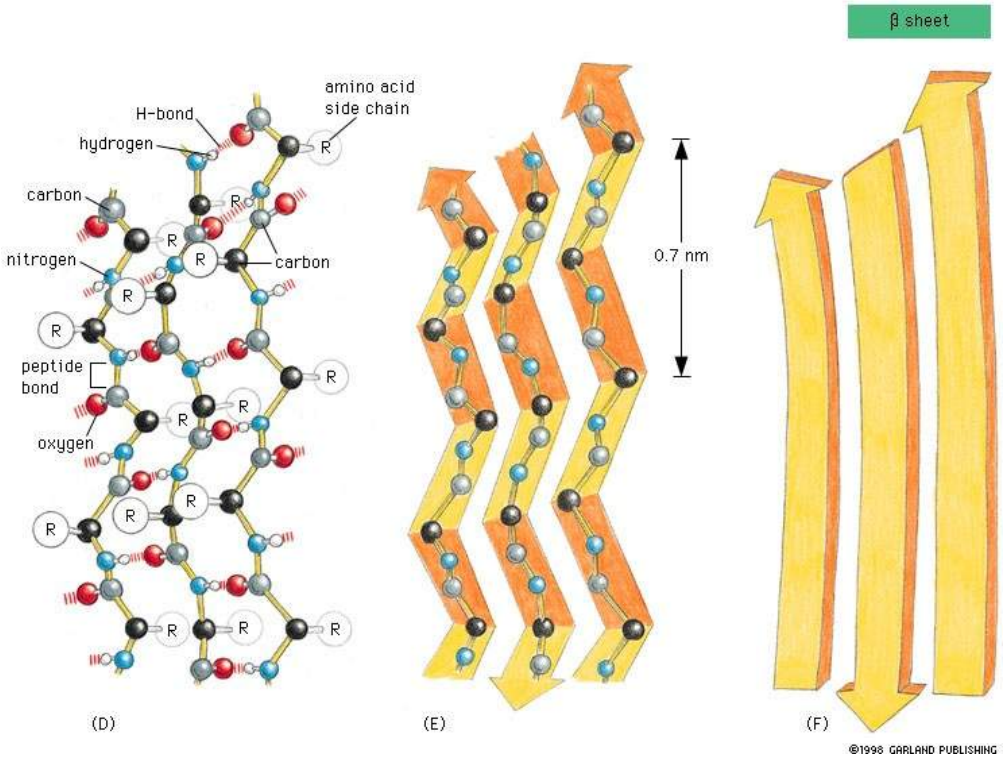
- The linear sequence of the linked amino acids contains the information necessary to generate a protein molecule with a unique three-dimensional shape
- The peptide bond allows for rotation around it and therefore the protein can fold and orient the R groups in favorable positions
- **Polar amino acids** (acidic, basic and neutral) are hydrophilic and tend to be placed on the **outside** of the protein.
- **Non-polar** (hydrophobic) amino acids tend to be placed on the **inside** of the protein

Types of interactions in protein folding

1. Hydrogen bond
2. Hydrophobic
3. Ionic
4. Disuphide linkage
5. VanderWaal's

Protein Folding





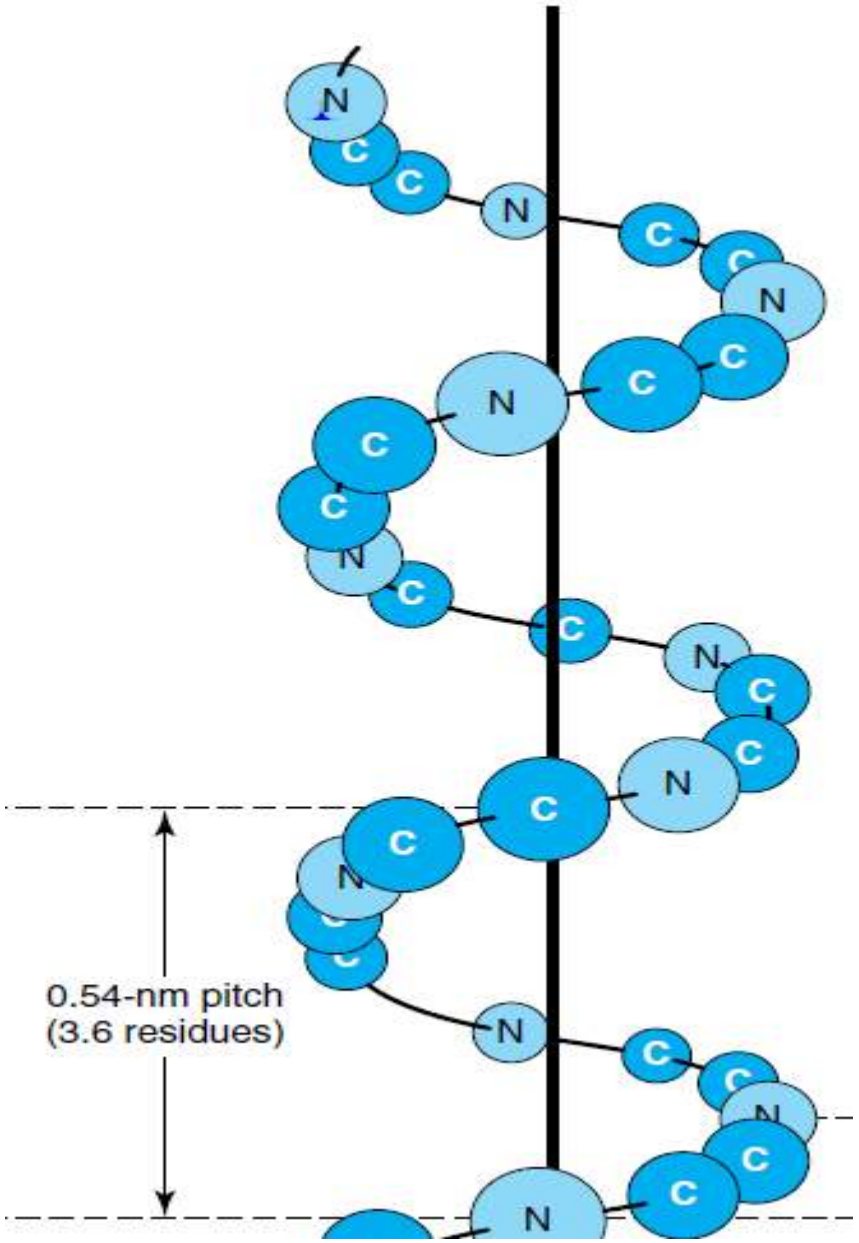
- 2 regular folding patterns have been identified – formed between the bonds of the peptide backbone
- α-helix – protein turns like a spiral – fibrous proteins (hair, nails, horns)
- β-sheet – protein folds back on itself as in a ribbon – globular protein

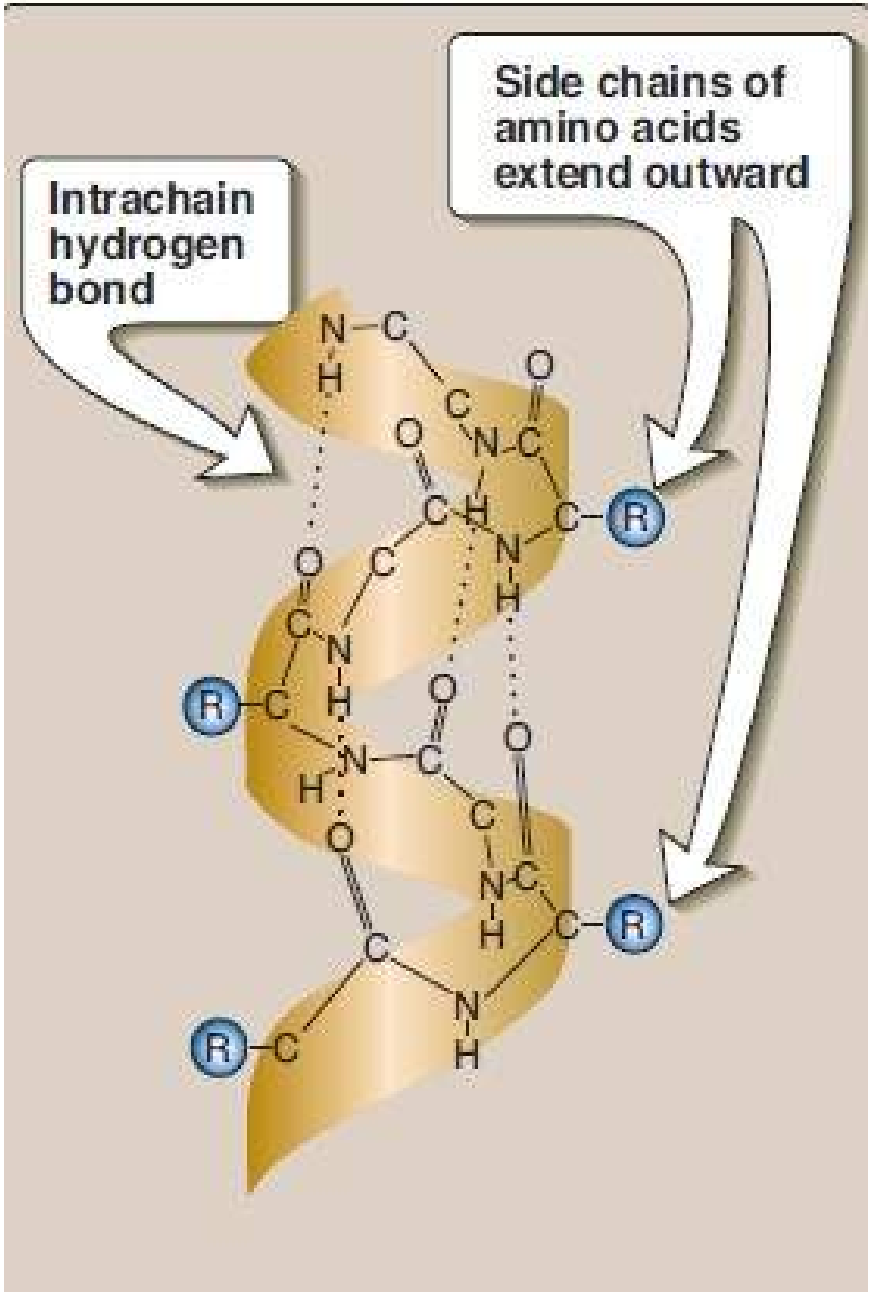
Secondary Structures

- The folding of short (3- to 30-residue), segments of polypeptide into geometrically ordered units
- Secondary structure refers to a local spatial arrangement of the polypeptide chain
- The **α helix**
  - stabilized by hydrogen bonds between nearby residues
- The **β sheet**
  - stabilized by hydrogen bonds between adjacent segments that may not be nearby
- Irregular arrangement of the polypeptide chain is called the random coil

The α helix

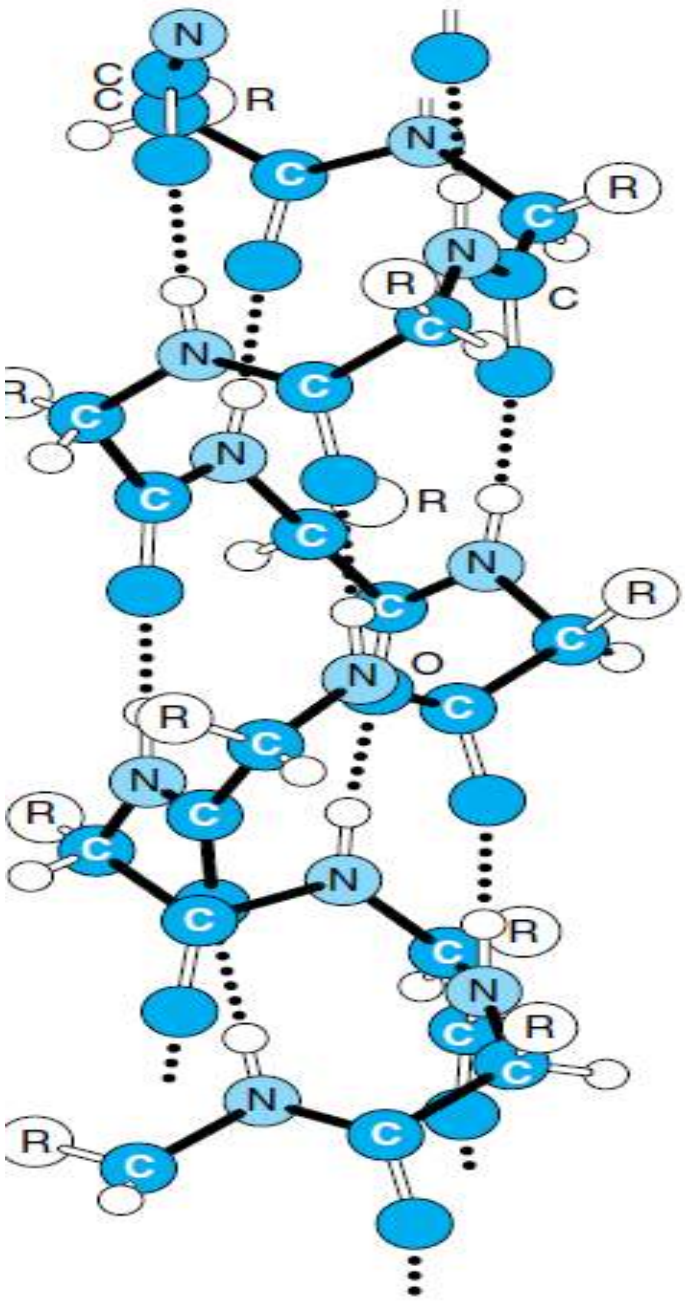
- A complete turn of the helix contains an average of 3.6 amino acyl residues (-C=O)
- The R group of each amino acyl residue in an alpha helix face outwards
- Proteins contain only L-amino acids, for which a right-handed α helix is by far more stable, and only right-handed α helices occur in nature

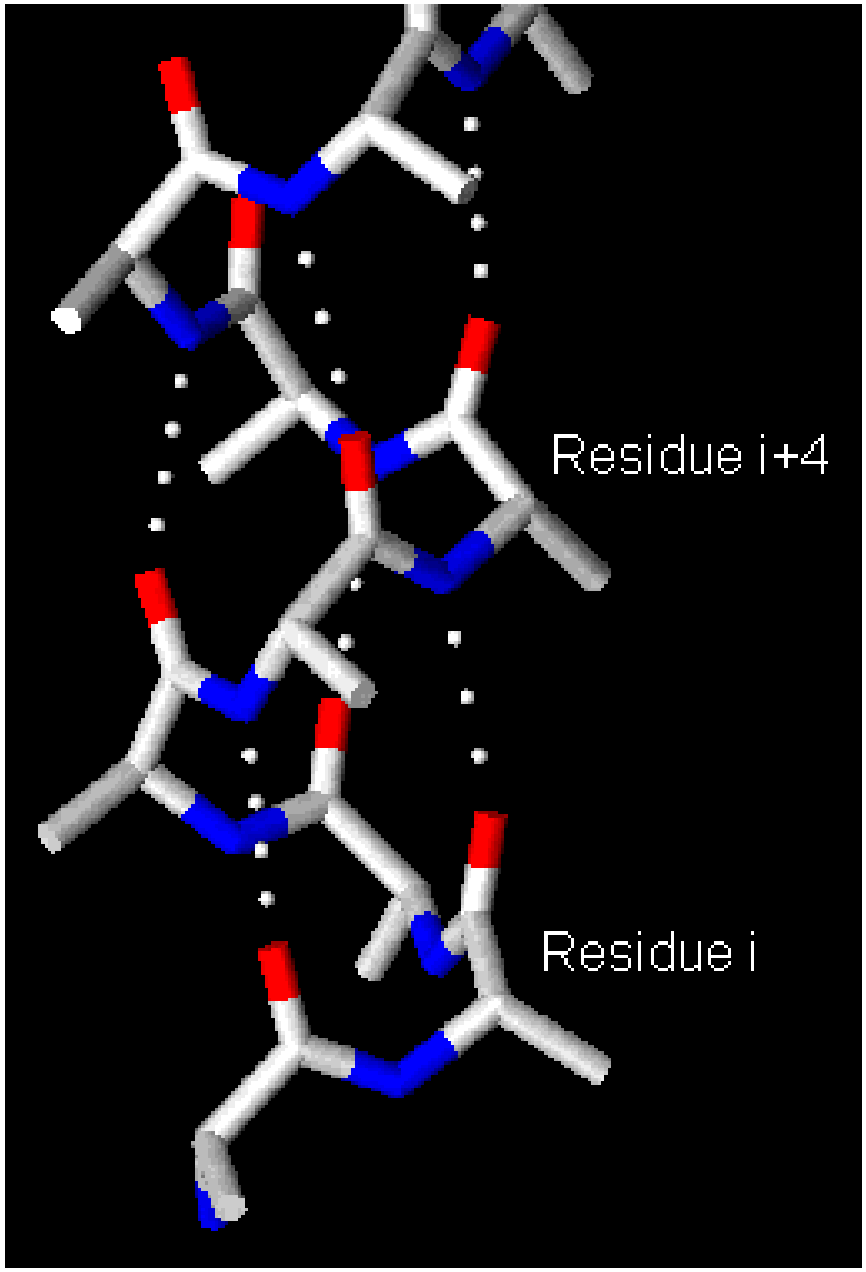

0.54-nm pitch
(3.6 residues)


Intrachain hydrogen bond
Side chains of amino acids extend outward

The α helix (stability)

- The stability of an α helix arises primarily from hydrogen bonds formed between the
  - oxygen of the peptide bond carbonyl group and the
  - hydrogen atom of the peptide bond nitrogen of the fourth residue down the polypeptide chain




Residue i+4
Residue i

Sequence Affects Helix Stability

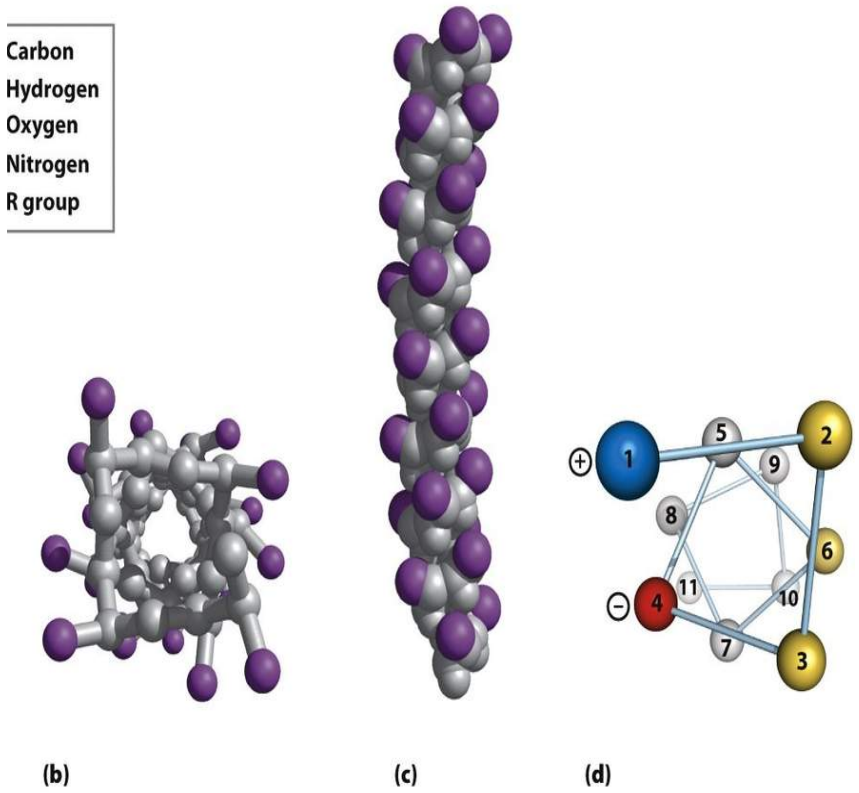
- Not all polypeptide sequences adopt α-helical structures
- Small hydrophobic residues such as Ala and Leu are strong helix formers
- Pro acts as a helix breaker because the rotation around the N-C_a bond is impossible
- Gly acts as a helix breaker because the tiny R-group supports other conformations

TABLE 4–1 Propensity of Amino Acids to Take Up an α-Helical Conformation

Amino acid	$\Delta\Delta G°$ (kJ/mol)*	Amino acid	$\Delta\Delta G°$ (kJ/mol)*
Ala	0	Leu	0.79
Arg	0.3	Lys	0.63
Asn	3	Met	0.88
Asp	2.5	Phe	2.0
Cys	3	Pro	>4
Gln	1.3	Ser	2.2
Glu	1.4	Thr	2.4
Gly	4.6	Tyr	2.0
His	2.6	Trp	2.0
Ile	1.4	Val	2.1

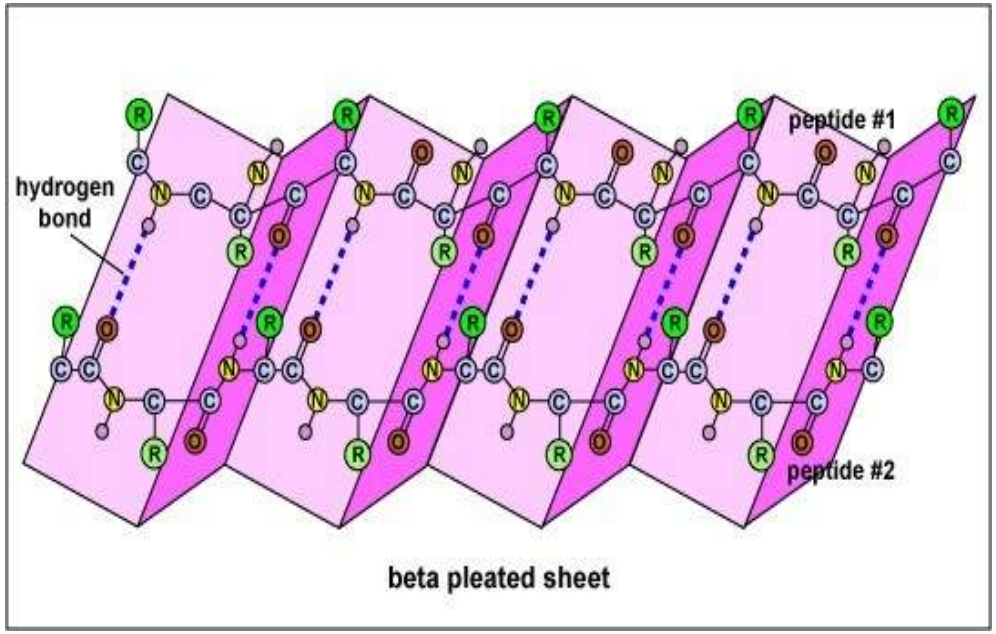
The amphipathic helix

- Many α helices have predominantly hydrophobic R groups on one side of the axis of the helix and *hydrophilic* ones on the other.
- can create a channel, or pore, that permits specific polar molecules to pass through hydrophobic cell membranes.

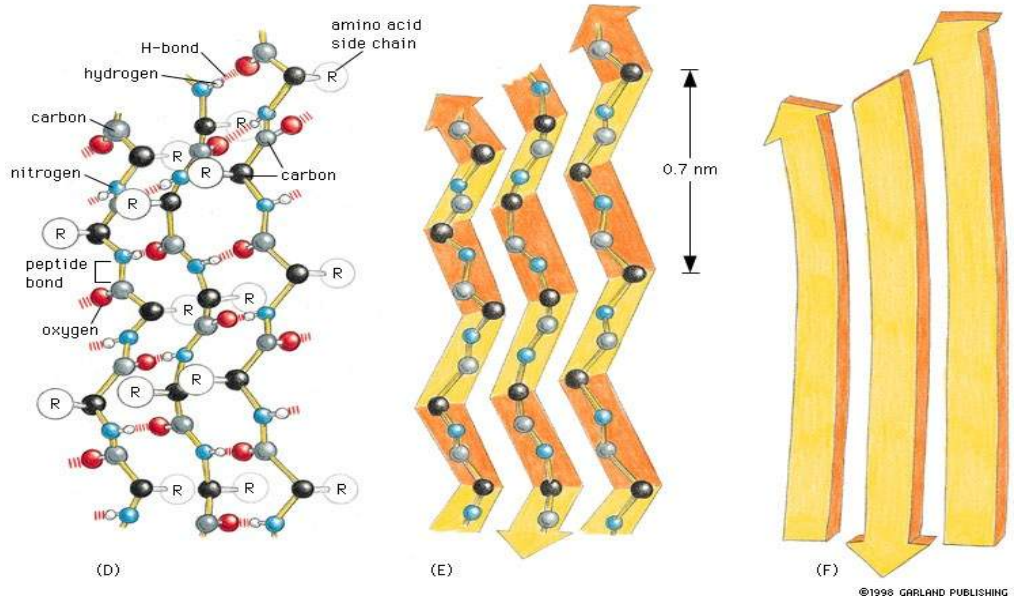


β-Sheet

- The β-sheet is another form of secondary structure in which all of the peptide bond components are involved in hydrogen bonding
- The surfaces of β-sheets appear “pleated,” and these structures are, therefore, often called “β-pleated sheets and visualized as broad arrows
- Core of many proteins is the β sheet
- Form rigid structures with the H-bond


hydrogen bond
peptide #1
peptide #2
beta pleated sheet

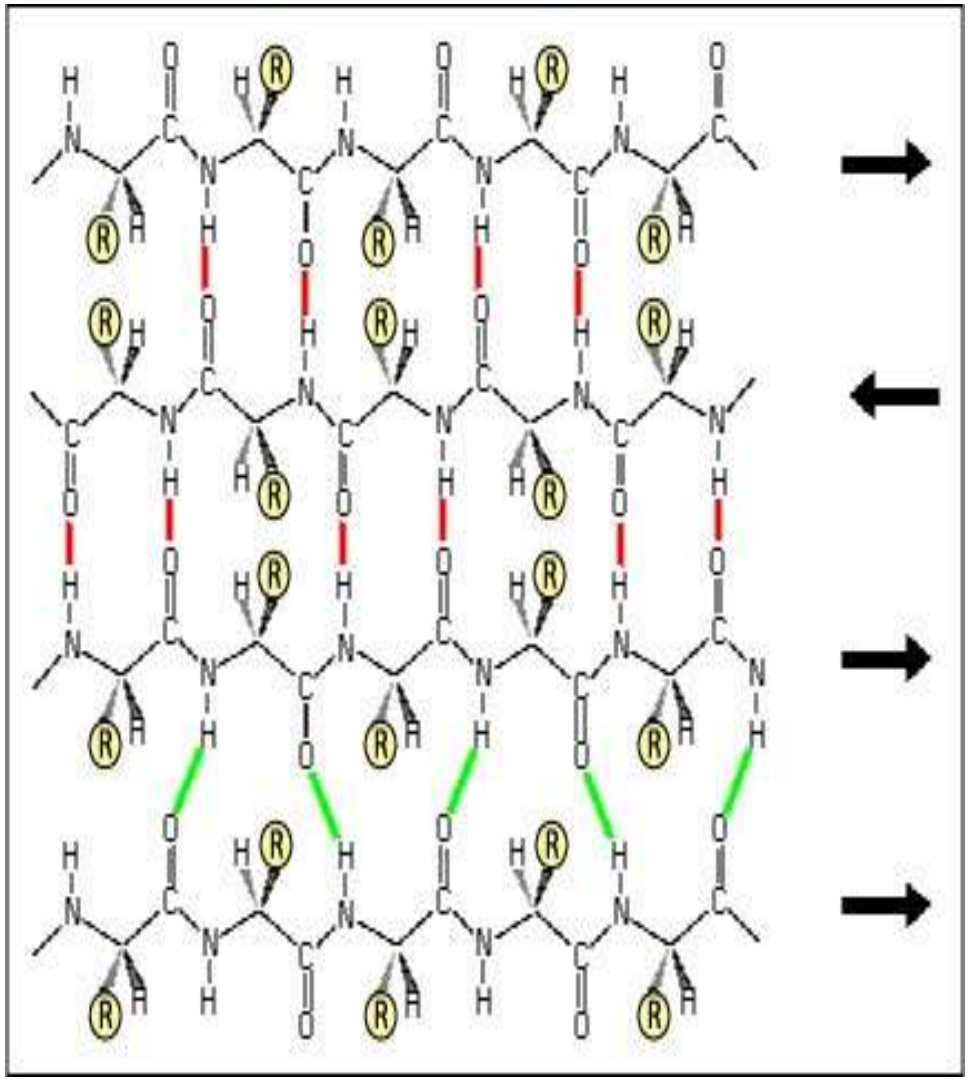
β sheet

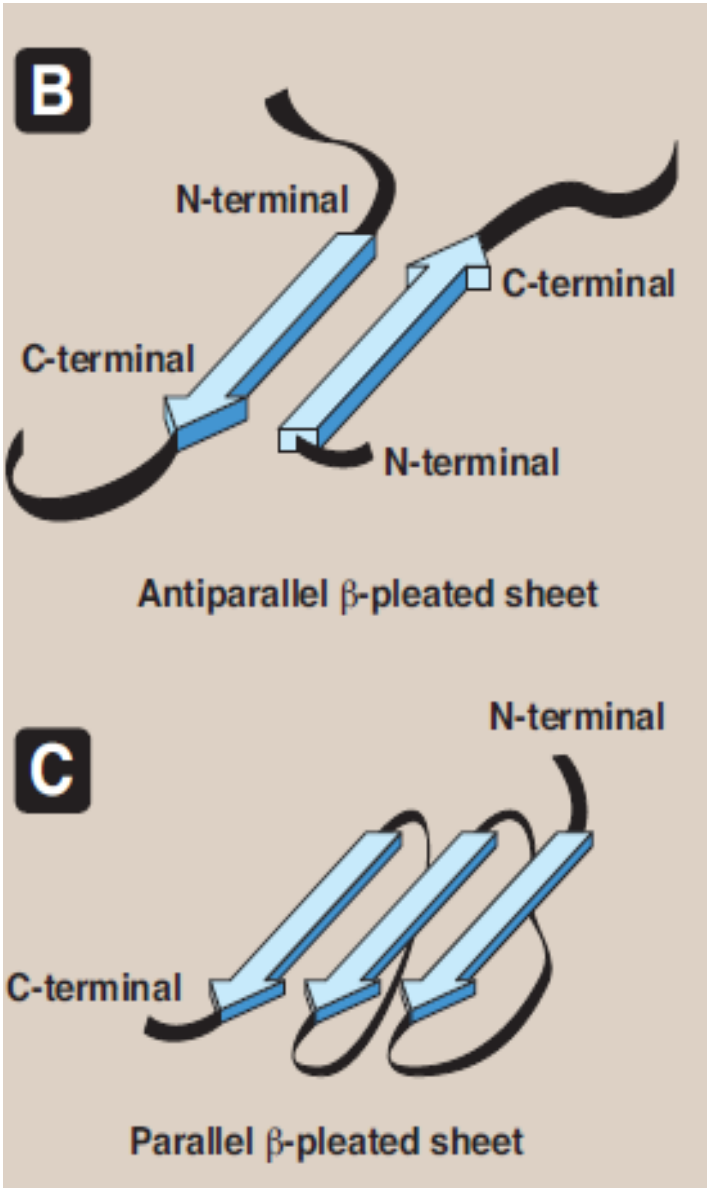

amino acid side chain
H-bond
hydrogen
carbon
nitrogen
peptide bond
oxygen
carbon
0.7 nm
(D)
(E)
(F)
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Parallel and antiparallel sheets

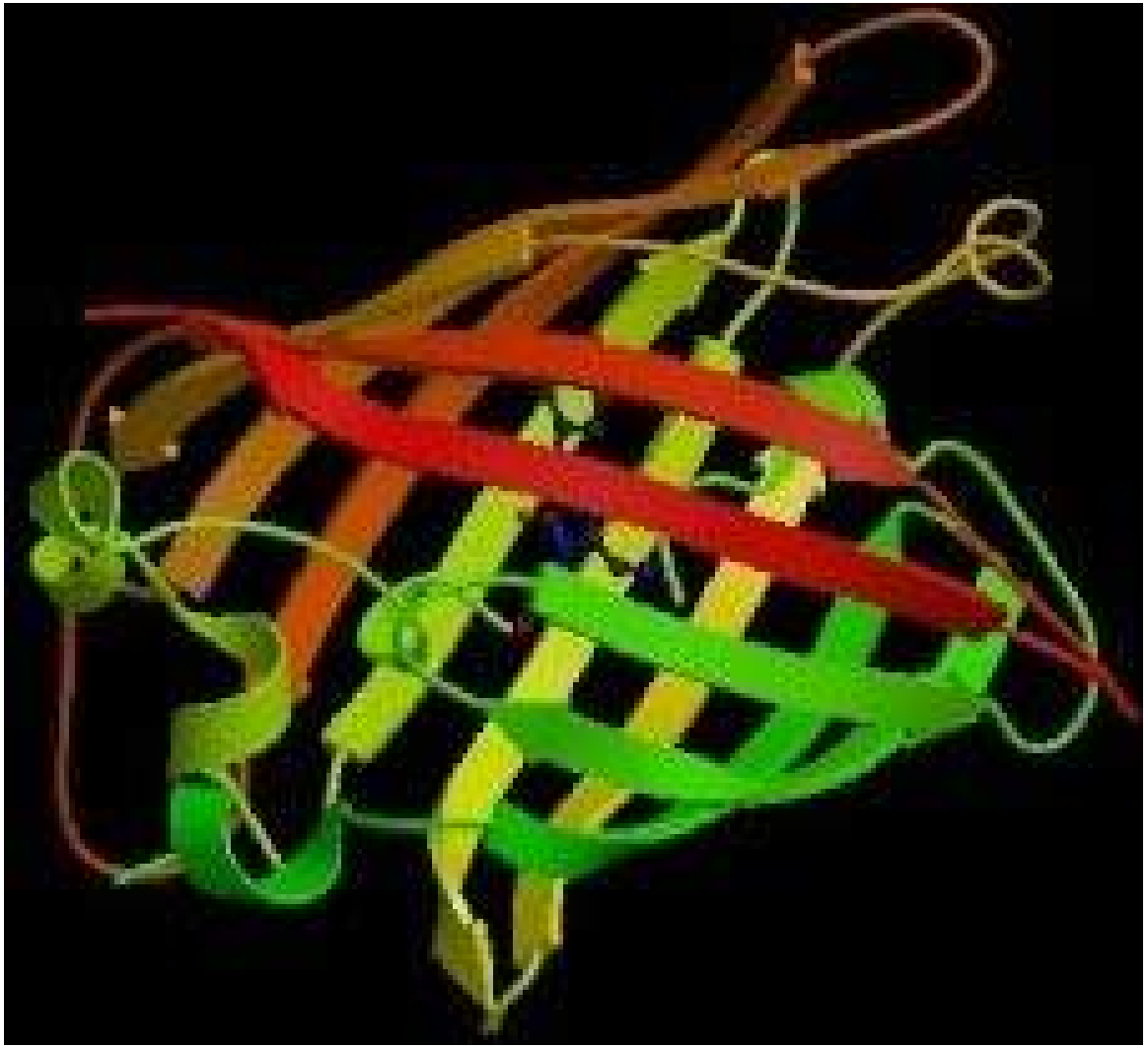
- A β-sheet can be formed from two or more separate polypeptide chains (interchain H-bond)or
- segments of same polypeptide chains that are arranged either antiparallel to each other (with the N-terminal and C-terminal ends of the β-strands alternating) or parallel to each other(with all the N-termini of the β-strands together).
- A β-sheet can also be formed by a single polypeptide chain folding back on itself

Protein structure: sheets (detail)





Clusters of twisted strands of β sheet folding upon itself forming the core of many globular proteins



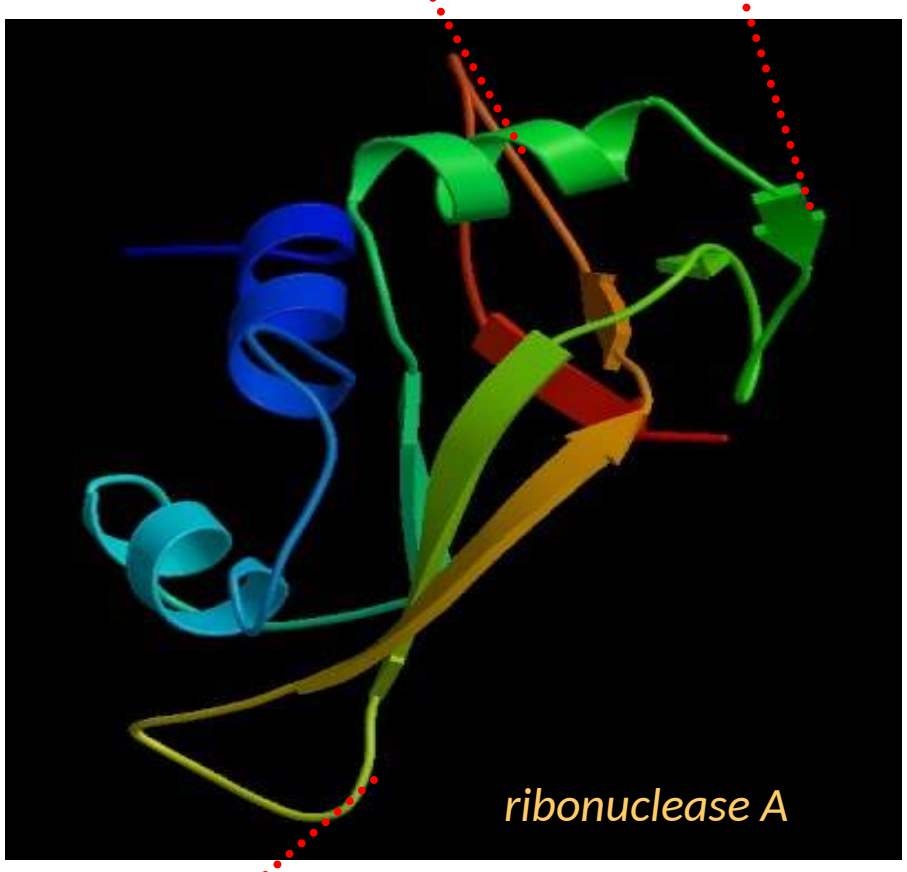
Loops & Bends

- Turns and bends refer to short segments of amino acids that join two units of secondary structure, such as two adjacent strands of an antiparallel β sheet.
- Irregular in conformation, loops nevertheless serve key biologic roles except in certain enzymes for binding the substrates.
- reside on the surface of proteins, thus are helpful in recognition and binding of antibodies.

Protein structure: turns/loops

- there are various types of turns, differing in the number of residues and H-bonding pattern

- loops are typically longer;
they are often called coils and do not have a ‘regular’,
or repeating, structure

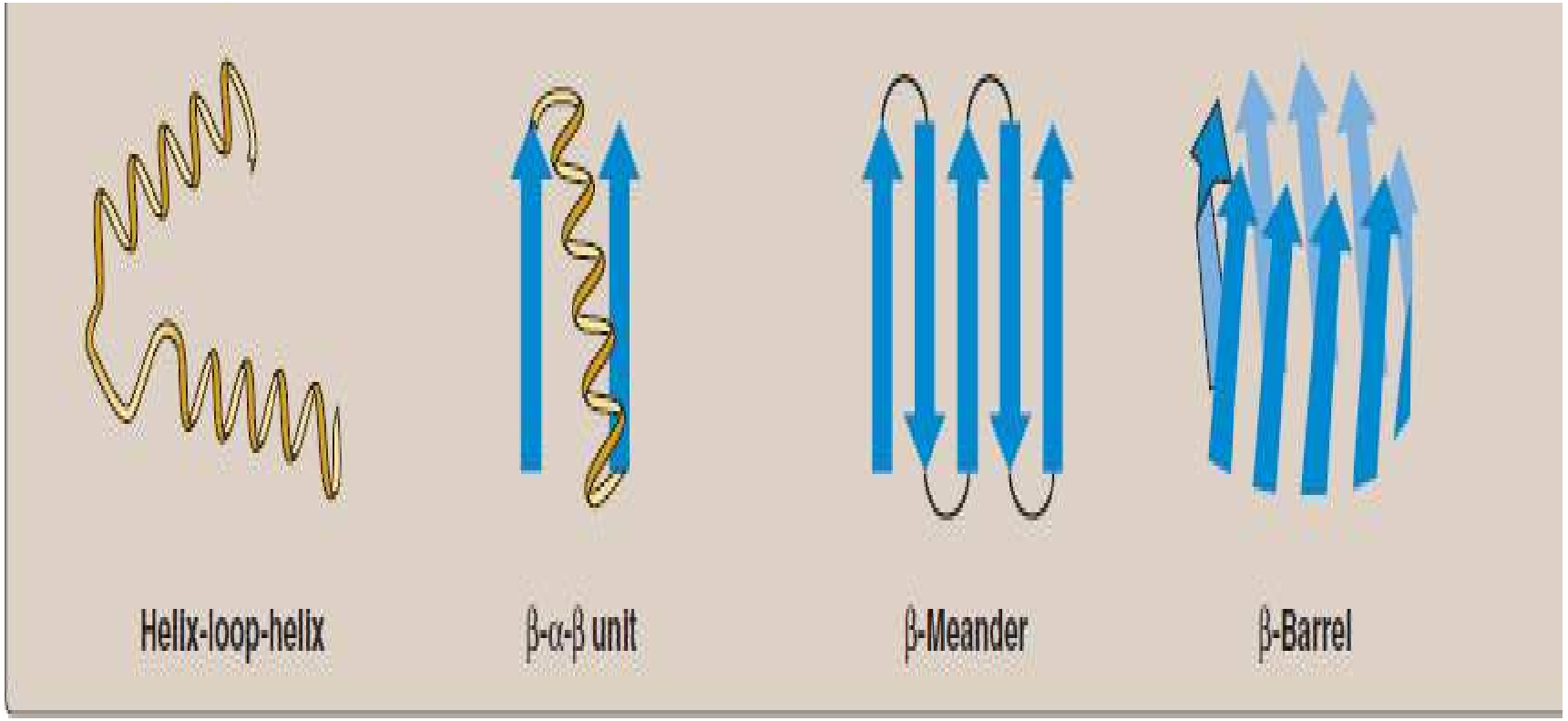


β-Bends (reverse turns, β-turns)

- β-Bends reverse the direction of a polypeptide chain, helping it form a compact, globular shape.
- β-Bends were given this name because they often connect successive strands of antiparallel β-sheets
- β-Bends are stabilized by the formation of hydrogen and ionic bonds.

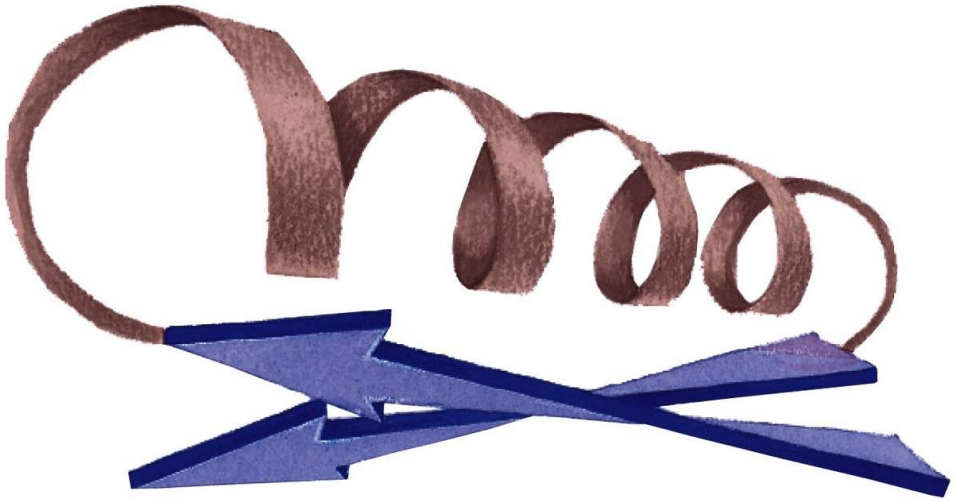
Supersecondary structures (motifs)

- Usually produced by packing side chains from adjacent secondary structural elements
- These form primarily the core region—that is, the interior of the molecule
- They are connected by loop regions (for example, β-bends) at the surface of the protein
- Thus, α-helices and β-sheets that are adjacent in the amino acid sequence are also usually(but not always) adjacent in the final, folded protein.


Helix-loop-helix
β-α-β unit
β-Meander
β-Barrel

Motifs (folds)

Arrangements of several secondary structure elements



β-α-β Loop

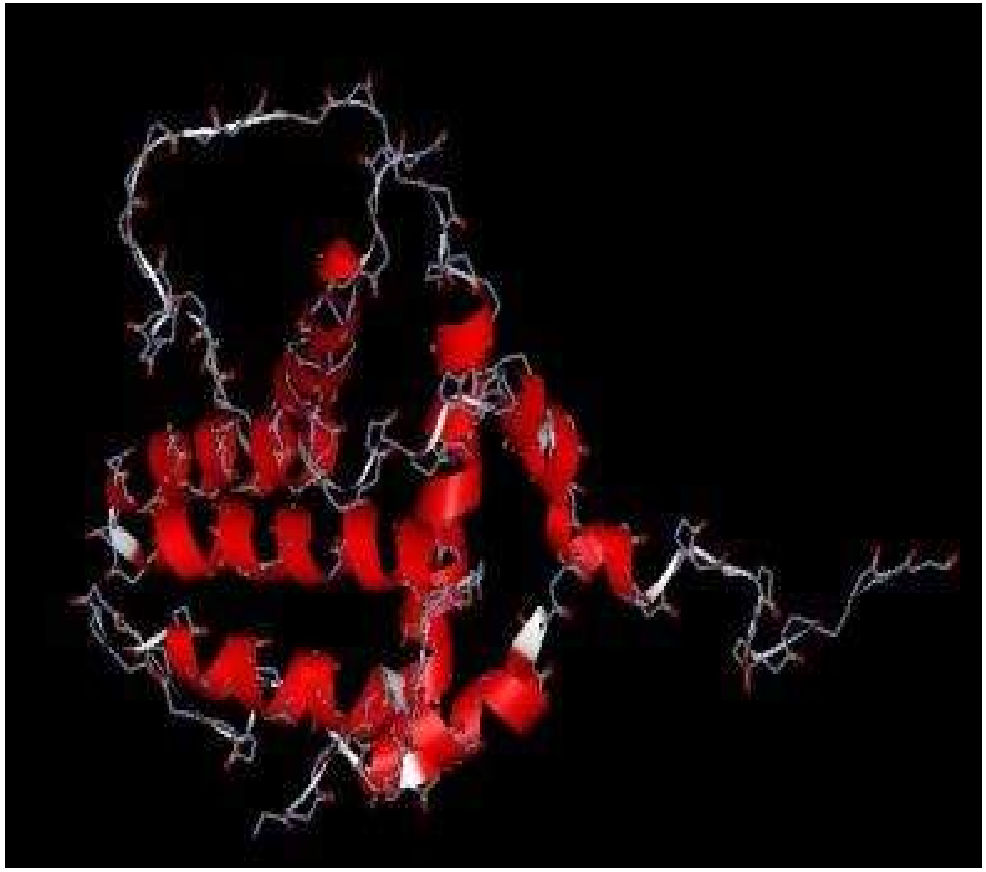
Figure 4-17a
Lehninger Principles of Biochemistry, Fifth Edition
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β Barrel

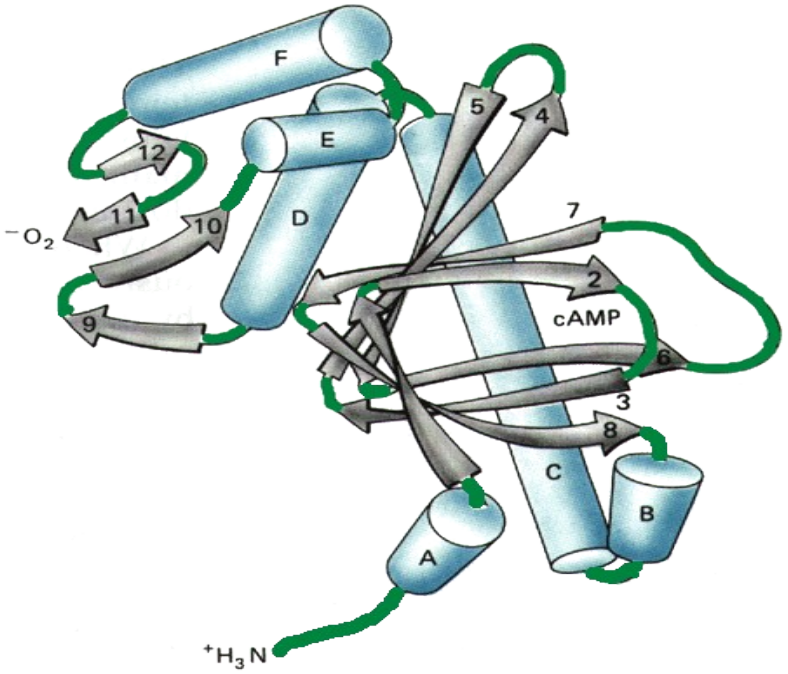
Figure 4-17b
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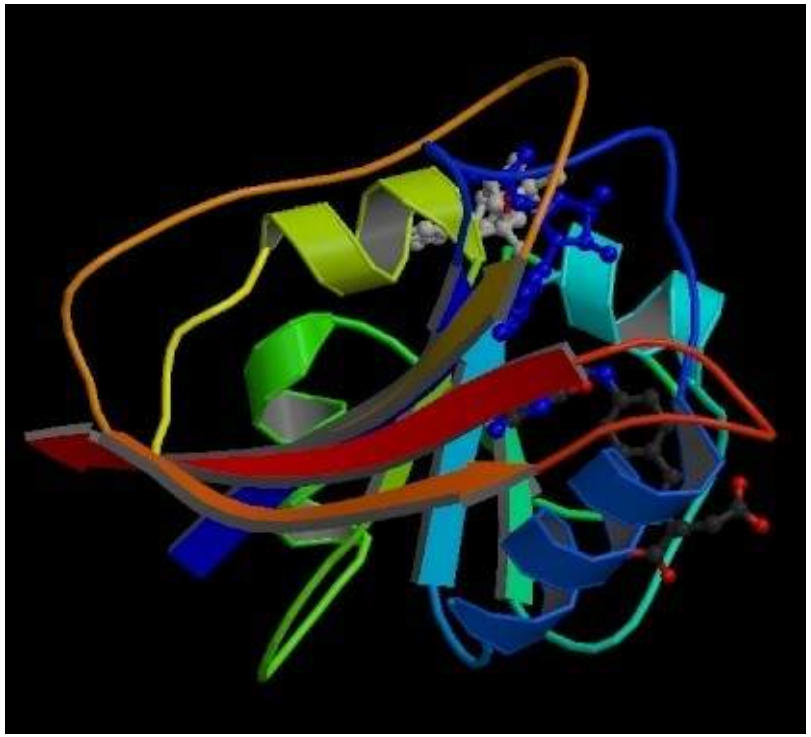
TERTIARY STRUCTURE OF GLOBULAR PROTEINS



Protein Tertiary Structure

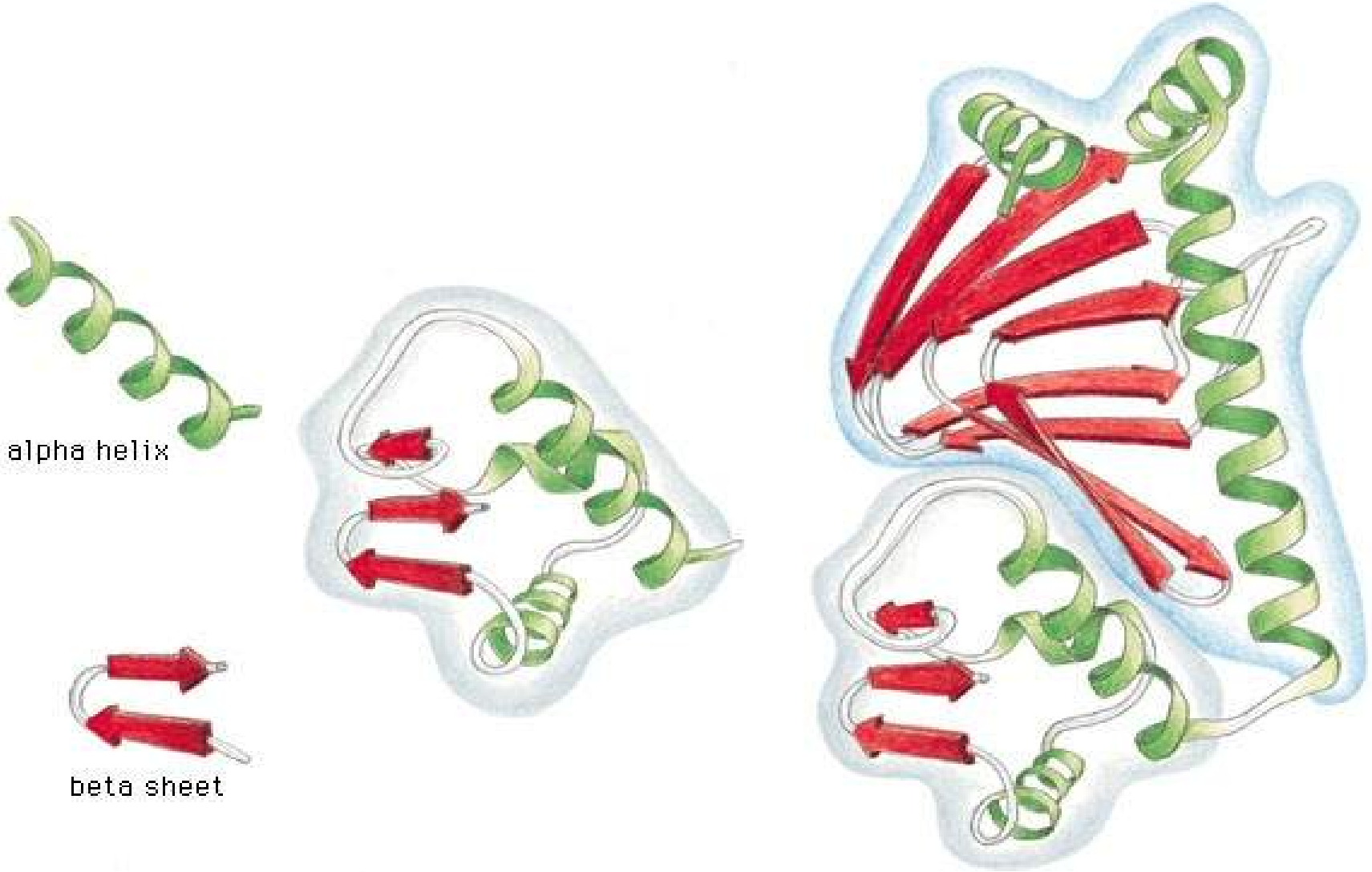
Refers to the entire three dimensional conformation of a polypeptide i.e. how secondary structural features —helices, sheets, bends, turns, and loops—assemble to form **domains**



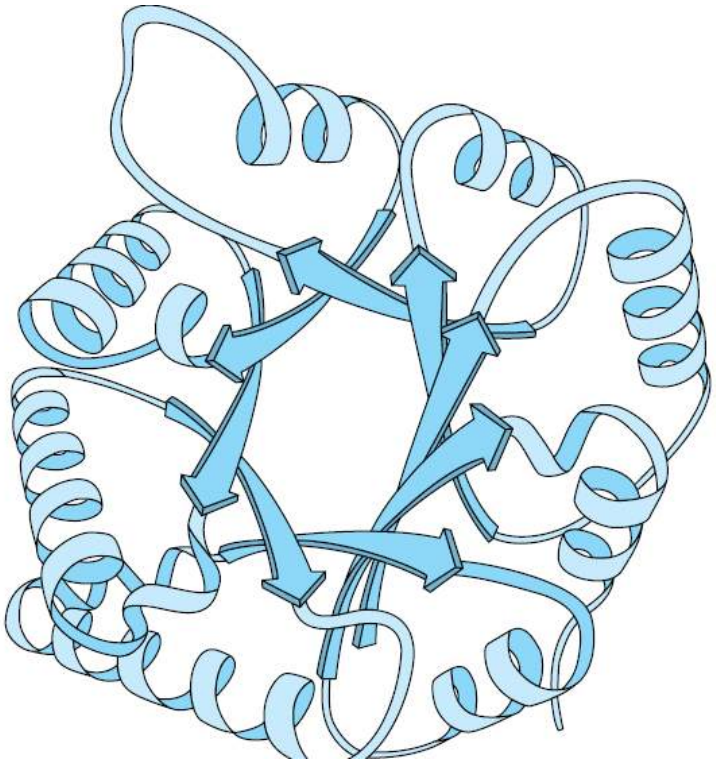


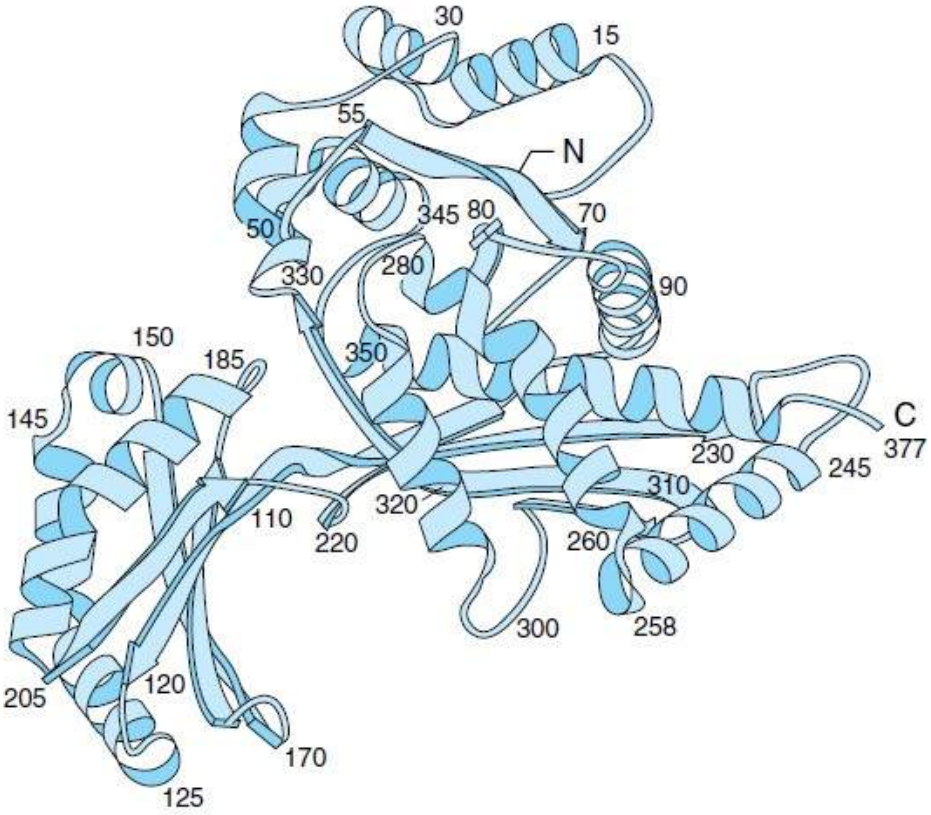
Domains

- A domain is a basic structural unit of a protein –a section of protein structure sufficient to perform a particular chemical or physical task such as binding of a substrate
- Other domains may anchor a protein to a membrane or interact with a regulatory molecule that modulates its function.
- Different domains can impart different functions to proteins
- Proteins can have one to many domains depending on protein size; chains greater than 200 amino acids in length generally consist of two or more domains



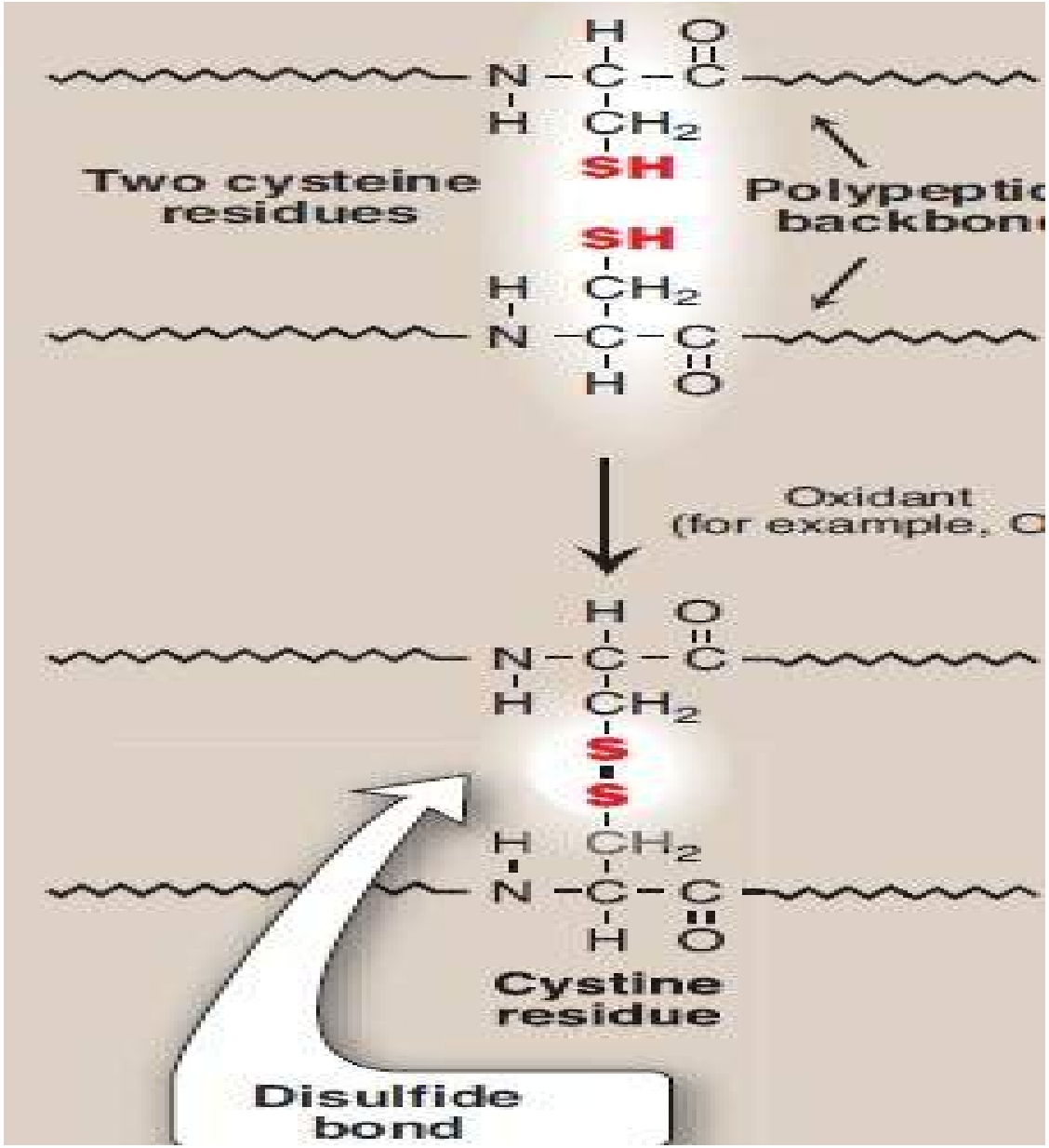
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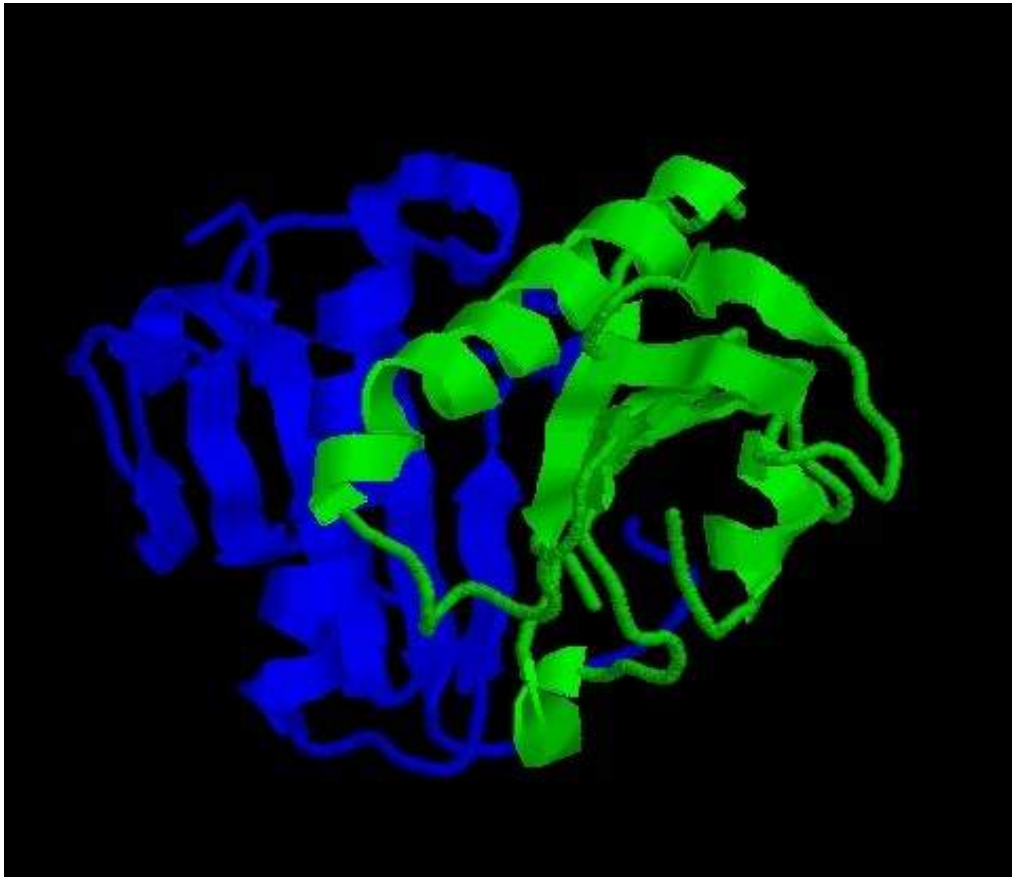
Interactions stabilizing tertiary structure

- Disulfide bonds (e.g. **cysteine-cysteine** disulphide bridge)
- Hydrophobic interactions (amino acids with nonpolar side chains tend to be located in the interior of the polypeptide molecule)
- Hydrogen bonds (amino acid side chains containing oxygen- or nitrogen-bound hydrogen)
- Ionic interactions (negatively charged groups, such as the carboxylate group (– COO–) can interact with positively charged groups, such as the amino group (– NH3+)


Two cysteine residues
Polypeptic backbon
Oxidant (for example, C
Cystine residue
Disulfide bond

QUATERNARY STRUCTURE OF PROTEINS

Quaternary structure defines the polypeptide composition of a protein and, for an oligomeric protein, the spatial relationships between its subunits or protomers.



- Many proteins consist of a single polypeptide chain. However, others may consist of two or more polypeptide chains that may be structurally identical or totally unrelated.
- The arrangement of these polypeptide subunits is called the quaternary structure of the protein.
- Subunits are held together by noncovalent interactions (for example, hydrogen bonds, ionic bonds, and hydrophobic interactions).

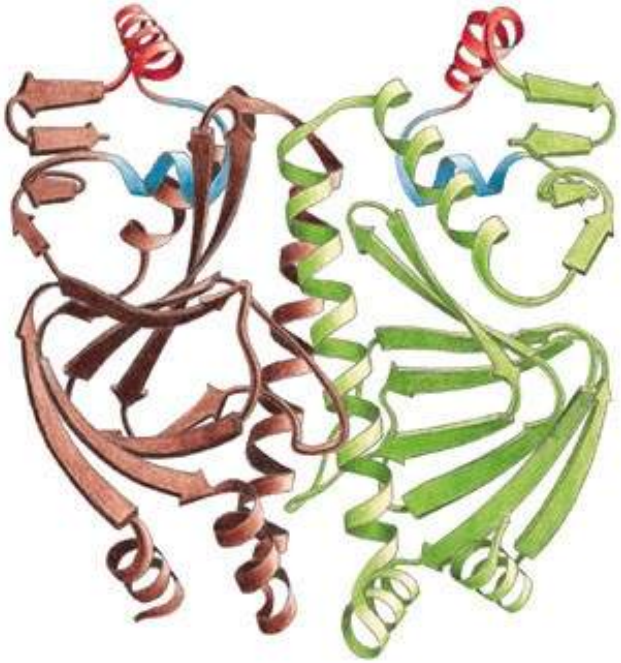
Proteins – Multiple Peptides

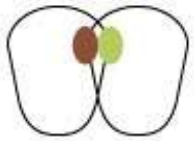
- **Monomeric proteins**

-consist of a single polypeptide chain.

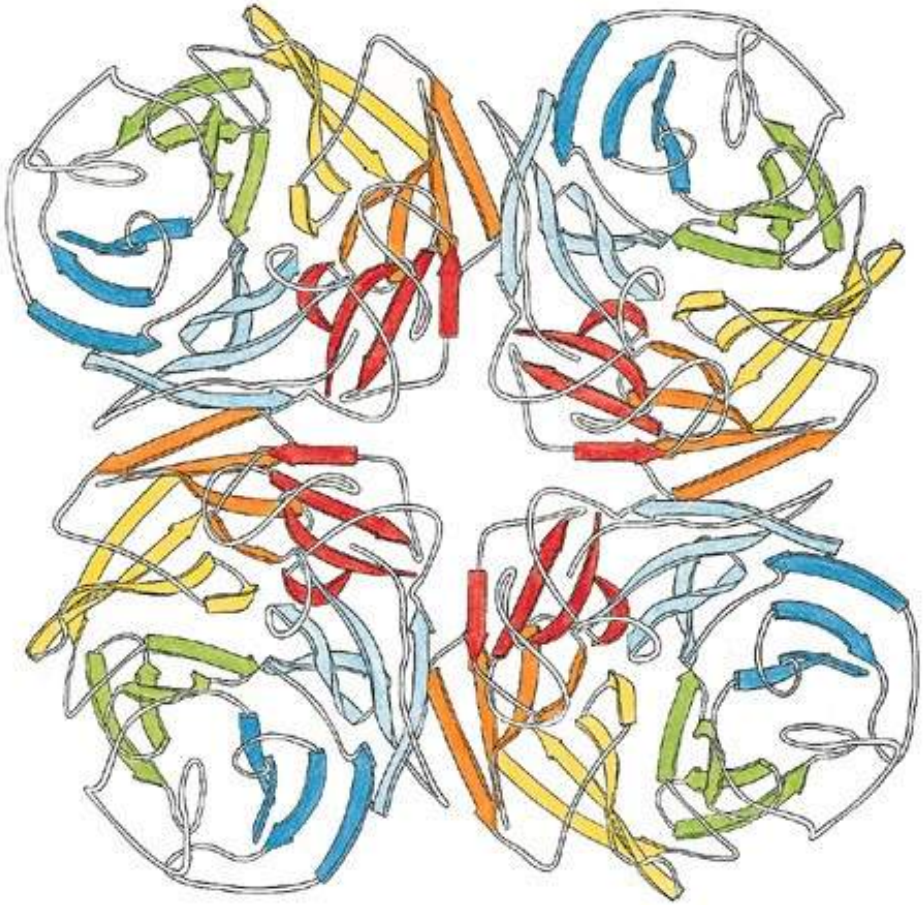
Dimeric proteins -contain two polypeptide chains.

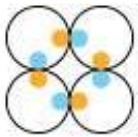
- **Homodimers -**contain two copies of the same polypeptide chain eg: α α or α_2
- **Heterodimer** the polypeptides differ.
- Greek letters (α, β, γ etc) are used to distinguish different subunits of a heterodimers and subscripts indicate the number of each subunit type
- For example, α_4 designates a homotetrameric protein, and $\alpha_2\beta_2\gamma$ a protein with five subunits of three different types.





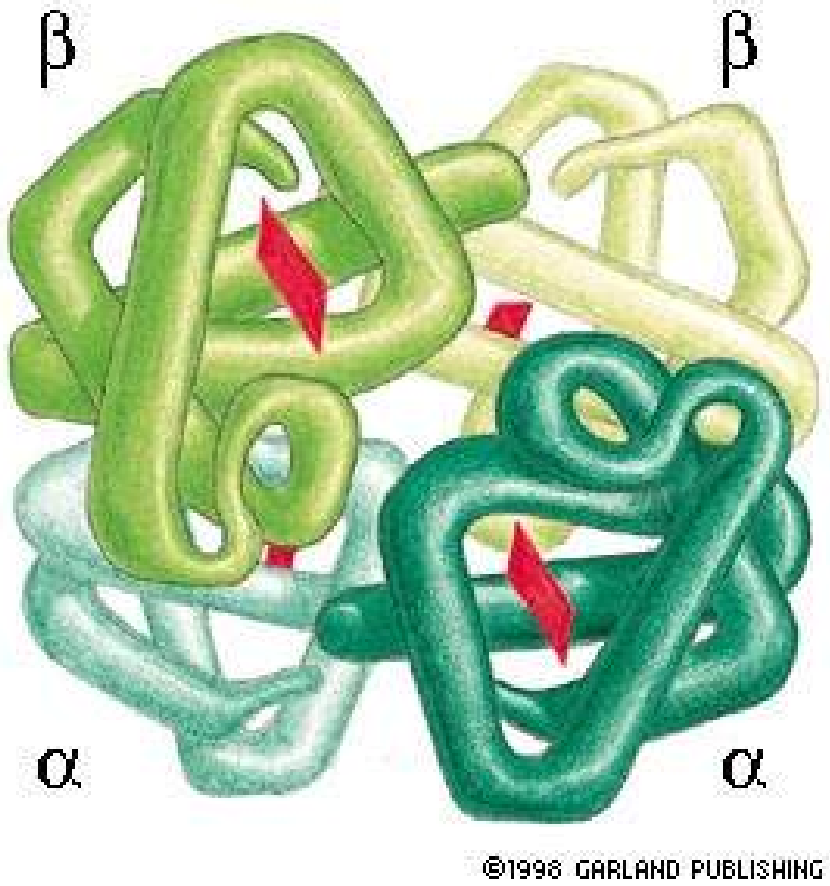
(A) dimer of the CAP protein





(B) tetramer of neuraminidase protein

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- Hemoglobin
  - 2 α globin subunits
  - 2 β globin subunits

Protein denaturation

- Protein denaturation results in the unfolding and disorganization of the protein's secondary and tertiary structures, which are **not** accompanied by hydrolysis of peptide bonds.
- Denaturing agents include heat, organic solvents, mechanical mixing, strong acids or bases, detergents, and ions of heavy metals such as lead and mercury.

Physico-chemical aspects

- Simple proteins
- Compound Proteins
- Derived proteins

Simple proteins

- Upon hydrolysis they yield amino acids only.

Examples:

ALBUMINS : Egg albumin, serum albumin

GLOBULIN : Tissue globulin, serum globulin

GLIADINS : wheat gliadin, hordein (barley) etc.

glutelins, albuminoids, histones, protamines etc.

Compound proteins

- Composed of simple proteins combined with a non-protein substance. The non-protein substance is called **prosthetic group** or **cofactor**.

Examples:

chromo-proteins: haemoglobin in which prosthetic group is iron

phospho-proteins: casein in milk: in which prosthetic group is phosphoric acid

nucleoproteins, glycoproteins etc.

Derived Proteins

- These are not naturally occurring proteins and are obtained from simple proteins by the action of enzymes and chemical agents. Example: denatured proteins and peptides. Further classified as:
- Primary protein derivatives
- Secondary protein derivatives

Primary protein derivatives

- Derivatives formed through hydrolytic changes which involve only slight alterations.

Examples:

- Proteans -insoluble products of the action of water, very dilute acids or enzymes
- Metaproteins — products of action of acids and alkalis

Secondary protein derivatives

- Products of a more extended cleavage of the protein molecule

Examples are peptones, proteoses and peptides

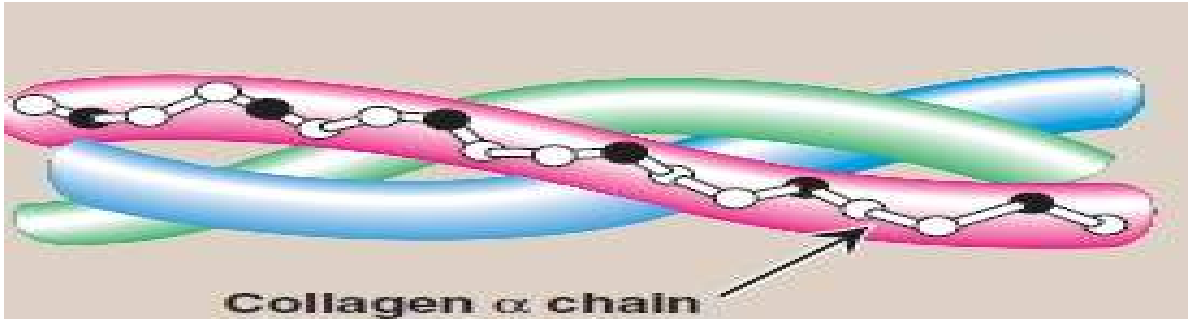
Classification by shape

Types of Proteins

- Globular Proteins – most of what we have dealt with so far
  - Compact shape like a ball with irregular surfaces
  - Enzymes are globular
- Fibrous Proteins – usually span a long distance in the cell
  - Its 3-D structure is usually long and rod shaped

Fibrous proteins

- Involved in structure: tendons ligaments blood clots
  (e.g. collagen and keratin)
- Contractile proteins in movement: muscle, microtubules
  (cytoskelton, mitotic spindle, cilia, flagella)



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Fibrous Proteins: From Structure to Function

Function	Structure	Example
Tough, rigid, hard (nails, horns)	Cross-linked α-helixes Rigid linker (S—S)	α-keratin
Tensile strength, non-stretching (tendons, cartilage)	Cross-linked triple-helixes Flexible linker (Lys-HyLys)	Collagen
Soft, flexible non-stretchy (egg sac, web)	Non-covalently held β-sheets van der Waals interaction	Silk fibroin

Structure of α-Keratin in Hair

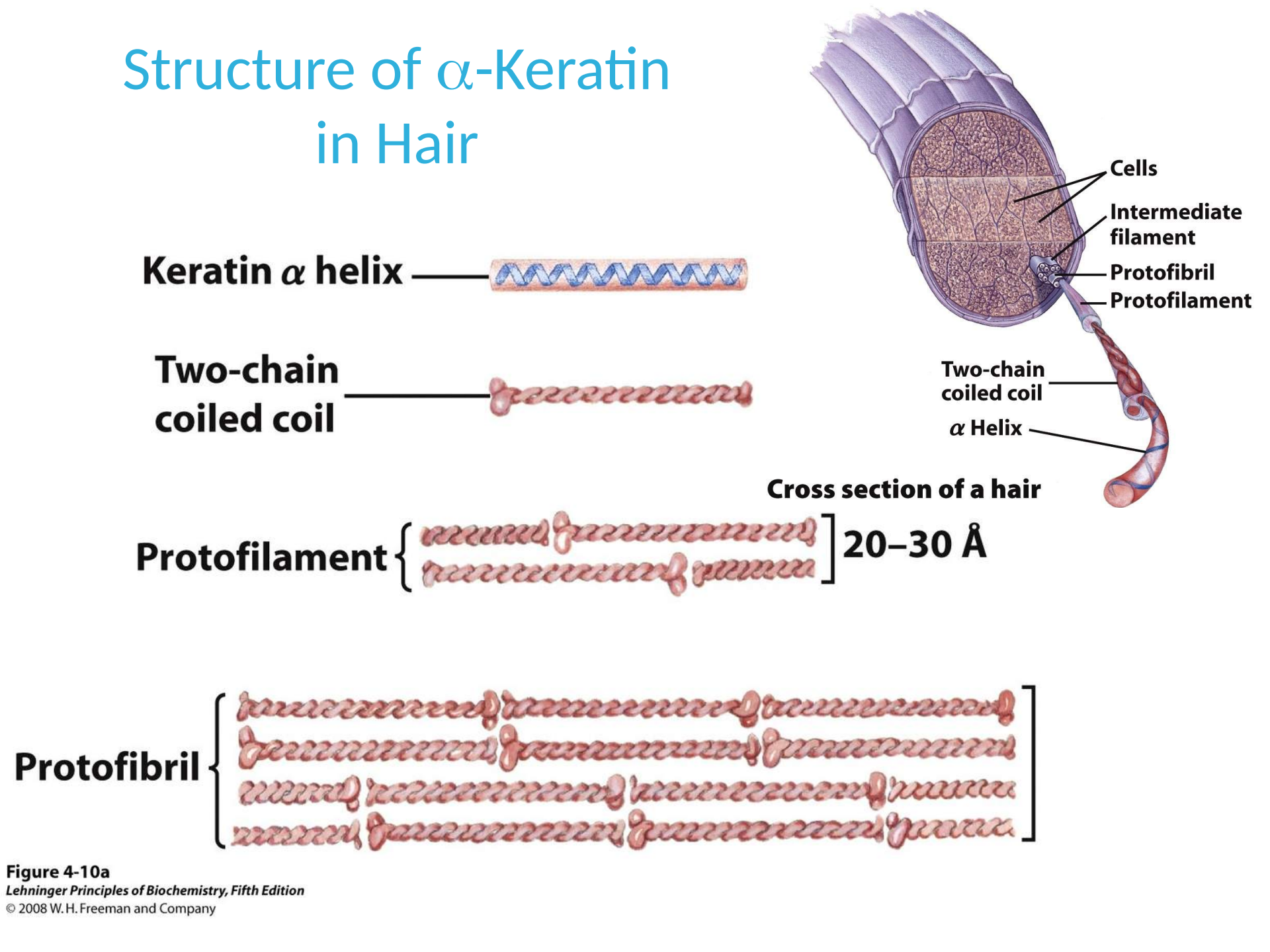
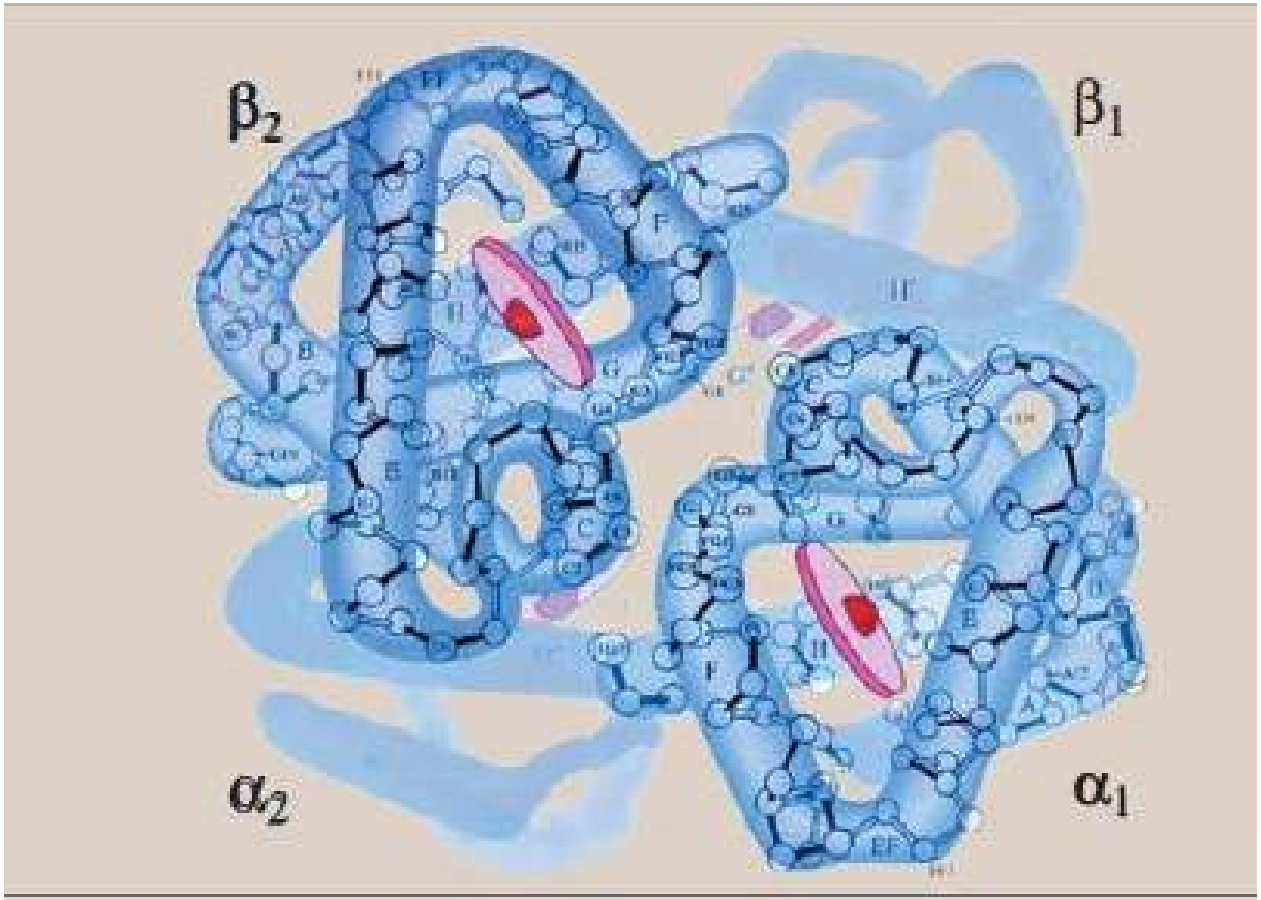


Figure 4-10a
Lehninger Principles of Biochemistry, Fifth Edition
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Globular proteins

- Compact ball shaped proteins
- Proteins with binding sites:
  enzymes, haemoglobin, immunoglobulins,
  membrane receptor sites

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β2
β1
α2
α1

Hierarchical nature of protein structure

Primary structure (Amino acid sequence)

↓

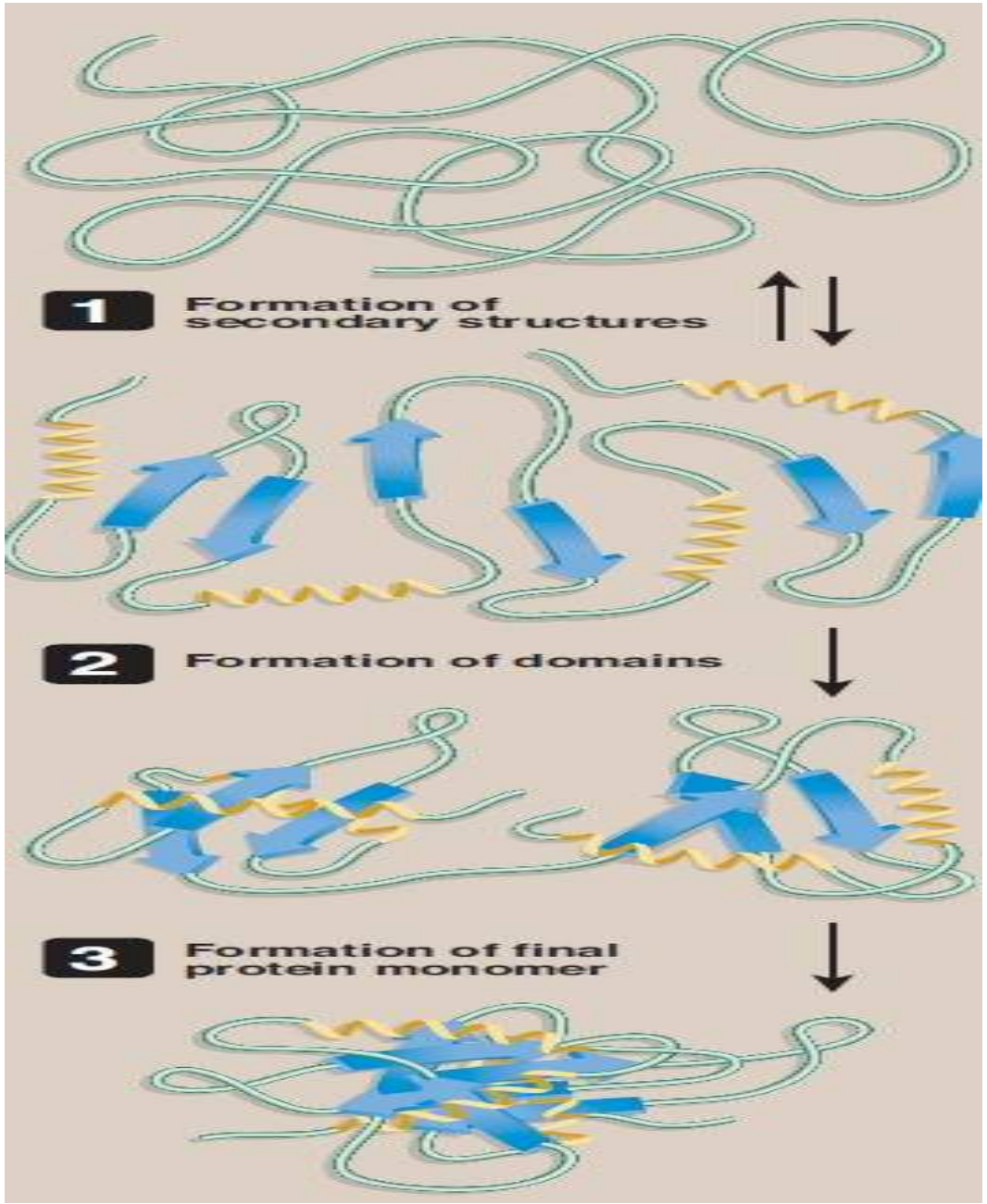
Secondary structure (α-helix, β-sheet)

↓

Tertiary structure (Three-dimensional structure formed by assembly of secondary structures)

↓

Quaternary structure (Structure formed by more than one polypeptide chains)

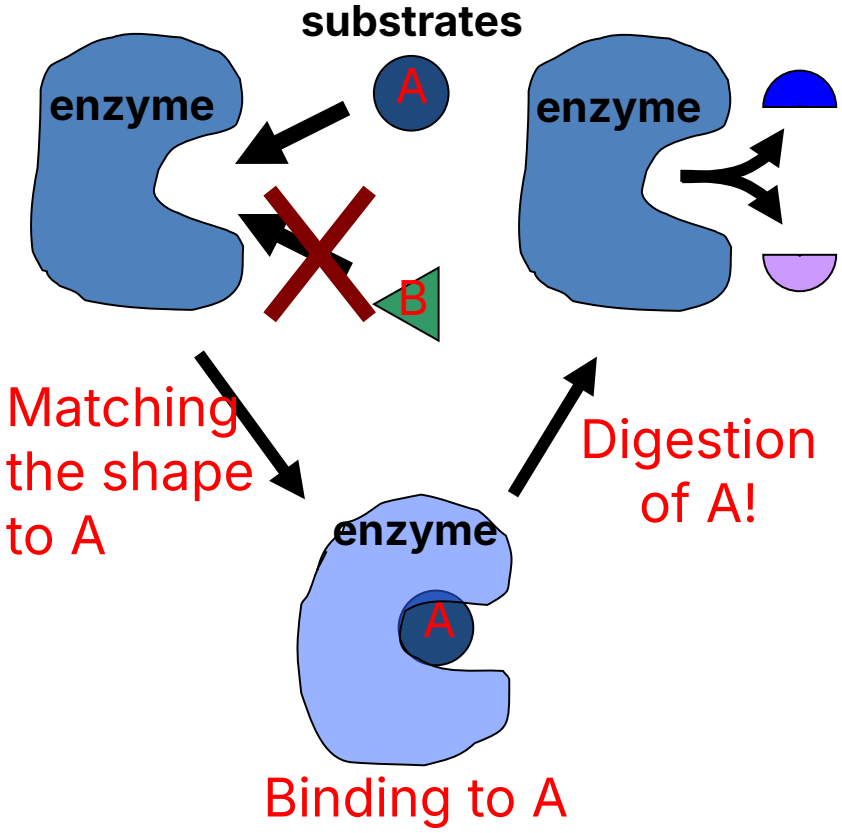

1
Formation of secondary structures
2
Formation of domains
3
Formation of final protein monomer

PROTEINS

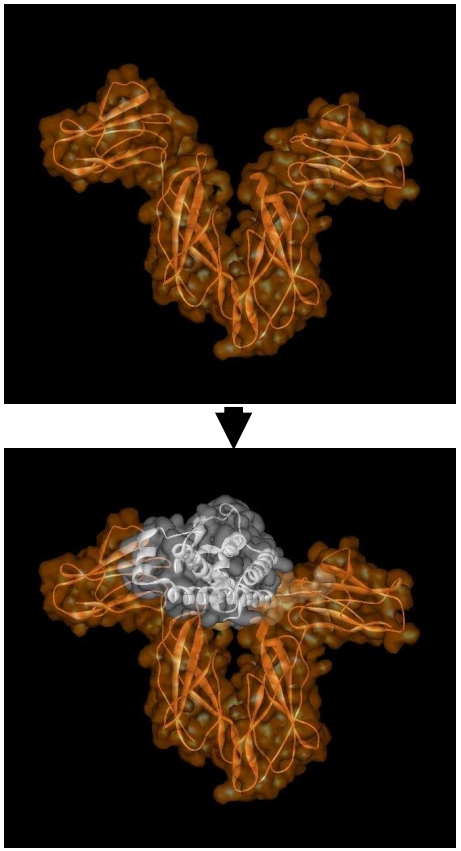
Structure Function relationship

Close relationship between protein structure and its function

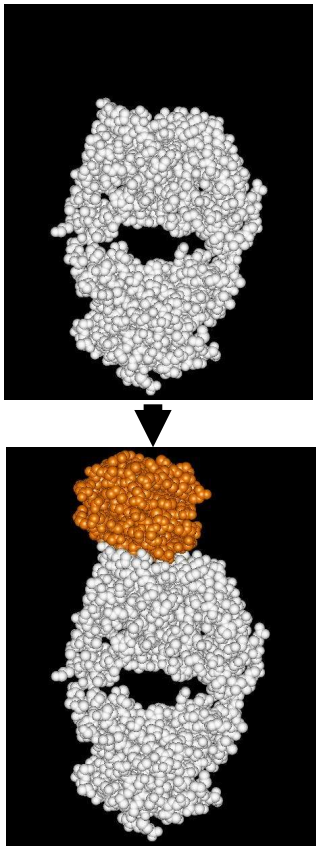
Example of enzyme reaction



Hormone receptor



Antibody



Proteins classified by function

The conformation of a protein gives it a unique function

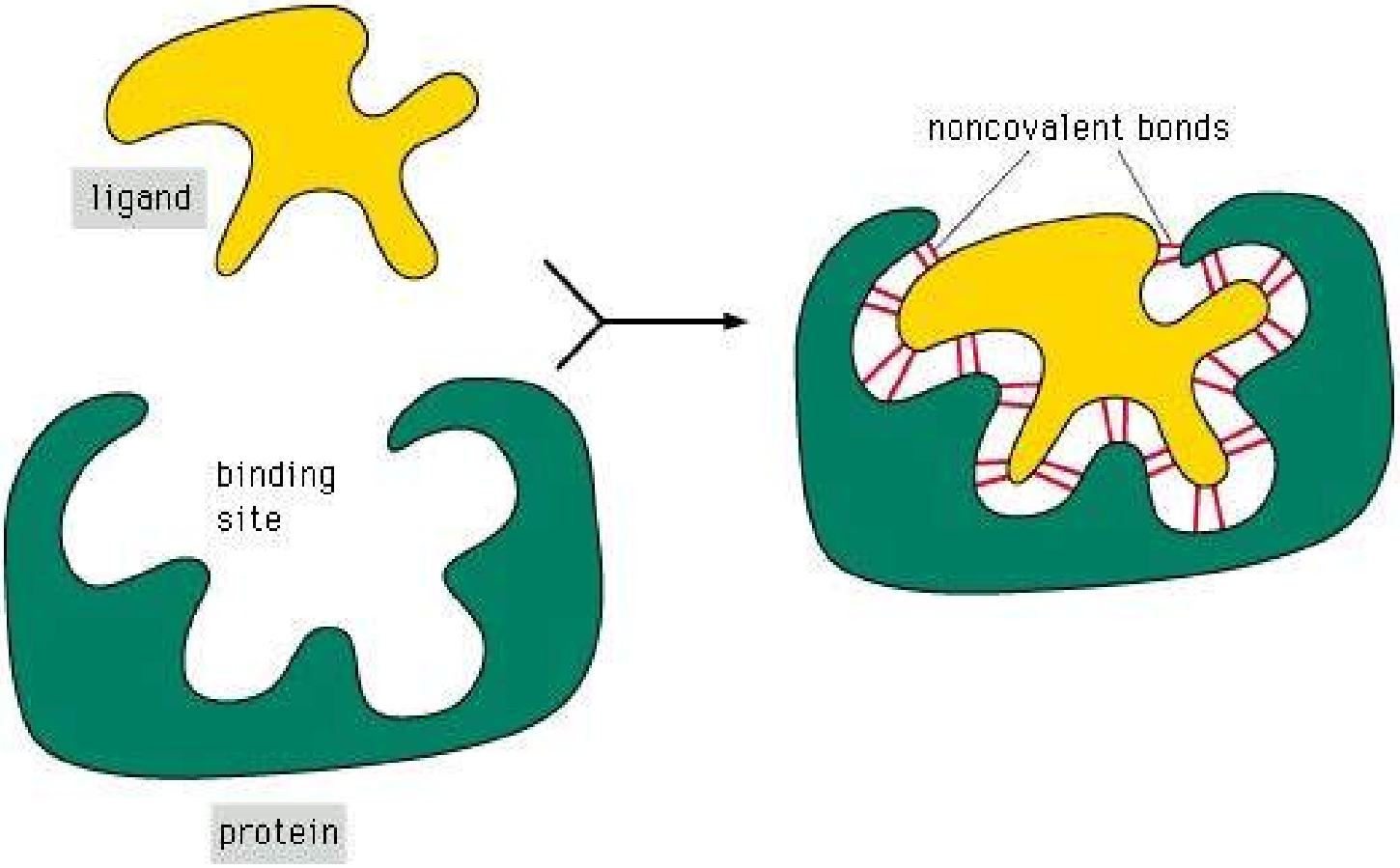
- **CATALYTIC:** enzymes
- **STORAGE:** ovalbumen (in eggs), casein (in milk)
- **TRANSPORT:** haemoglobin
- **COMMUNICATION:** hormones (eg insulin) and neurotransmitters
- **CONTRACTILE:** actin, myosin
- **PROTECTIVE:** Immunoglobulin, fibrinogen, blood clotting factors
- **TOXINS:** snake venom
- **STRUCTURAL:** cell membrane proteins, keratin (hair), collagen

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Enzymes as Catalysts

- Enzymes are proteins that bind to their ligand
- Output of the reaction is called the product
- Enzymes can repeat these steps many times rapidly and called **catalysts**

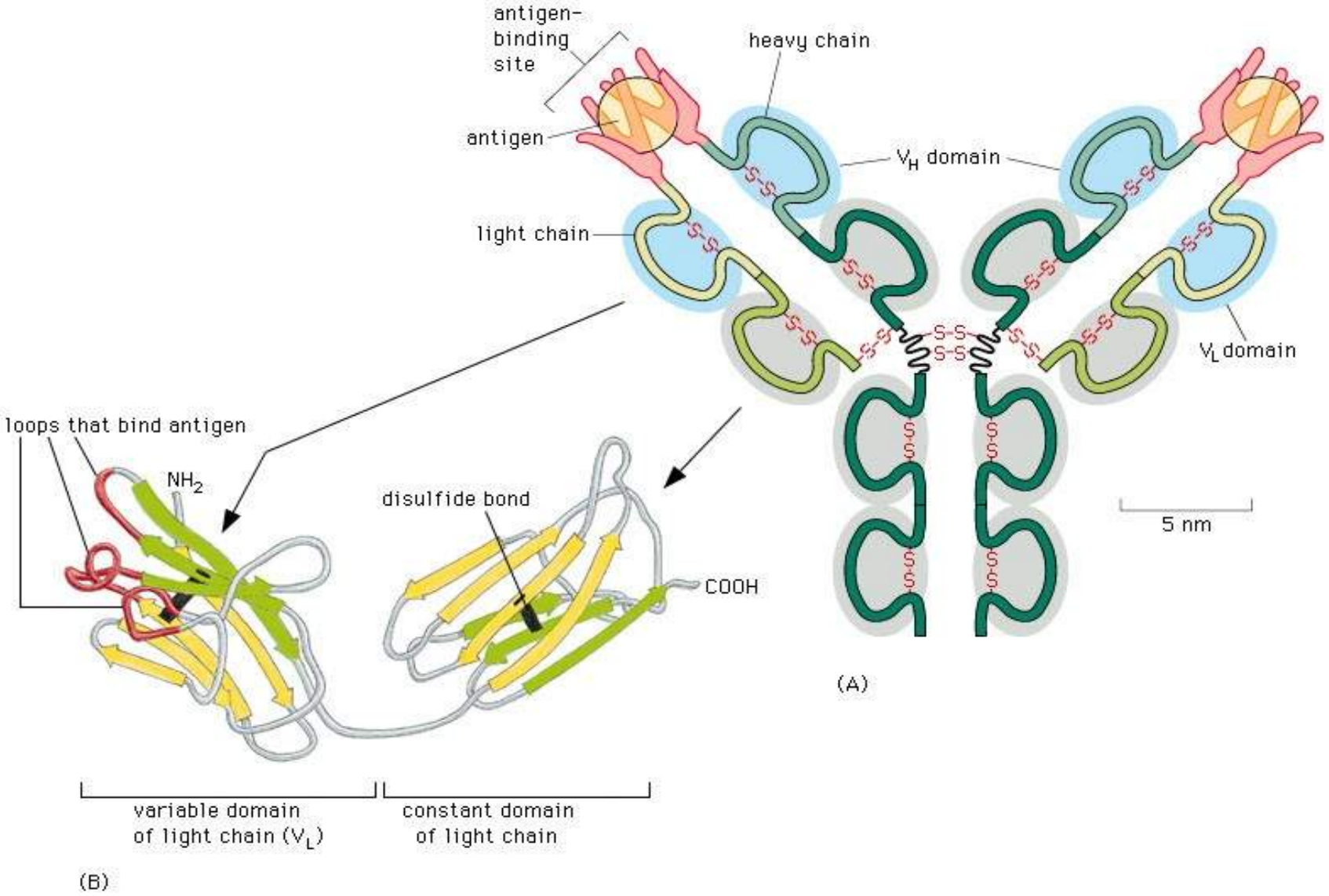
Ligand Binding



Antibody Family

- A family of proteins (immunoglobulins) are made in response to a foreign molecule ie. bacteria, virus, pollen… called the antigen
- Bind together tightly and therefore inactivates the antigen or marks it for destruction

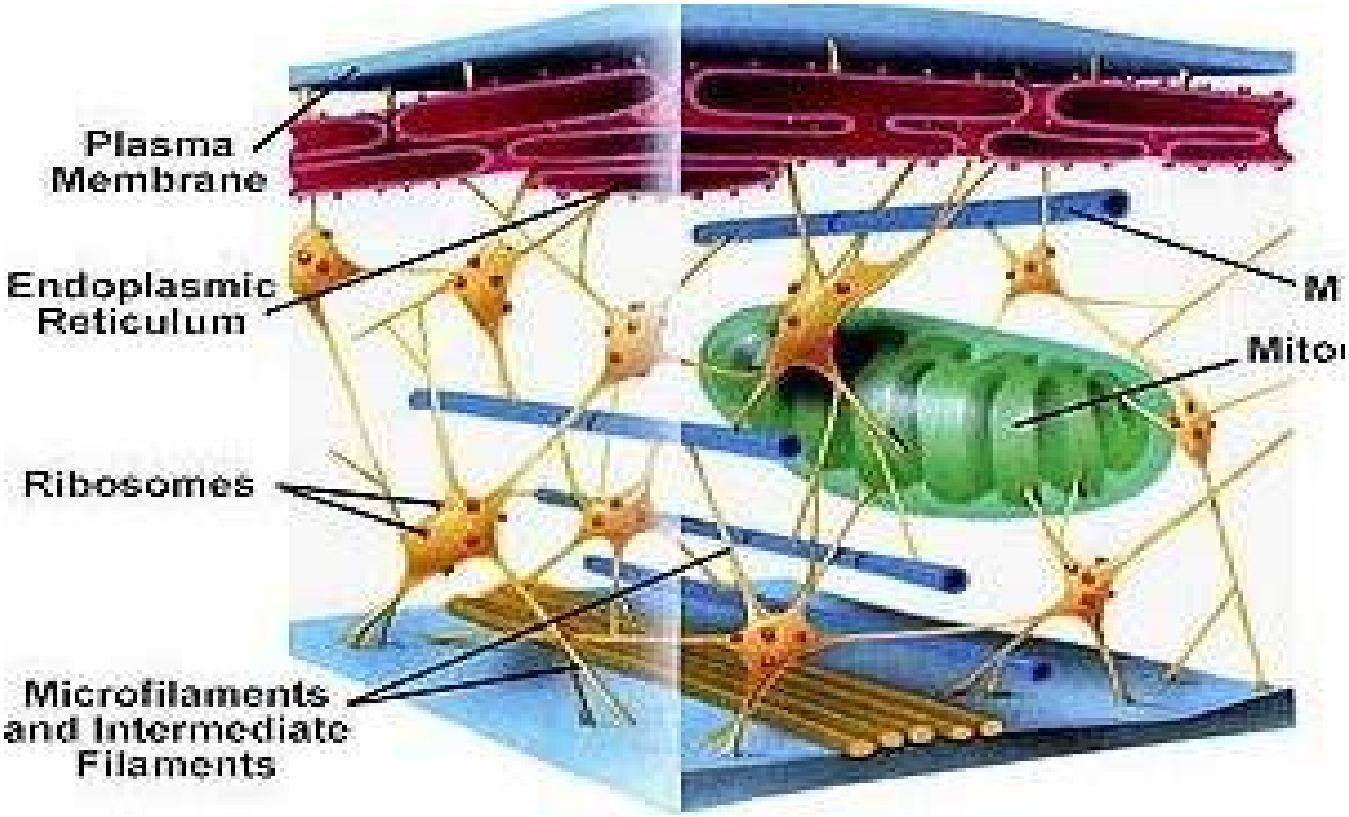
Antibodies

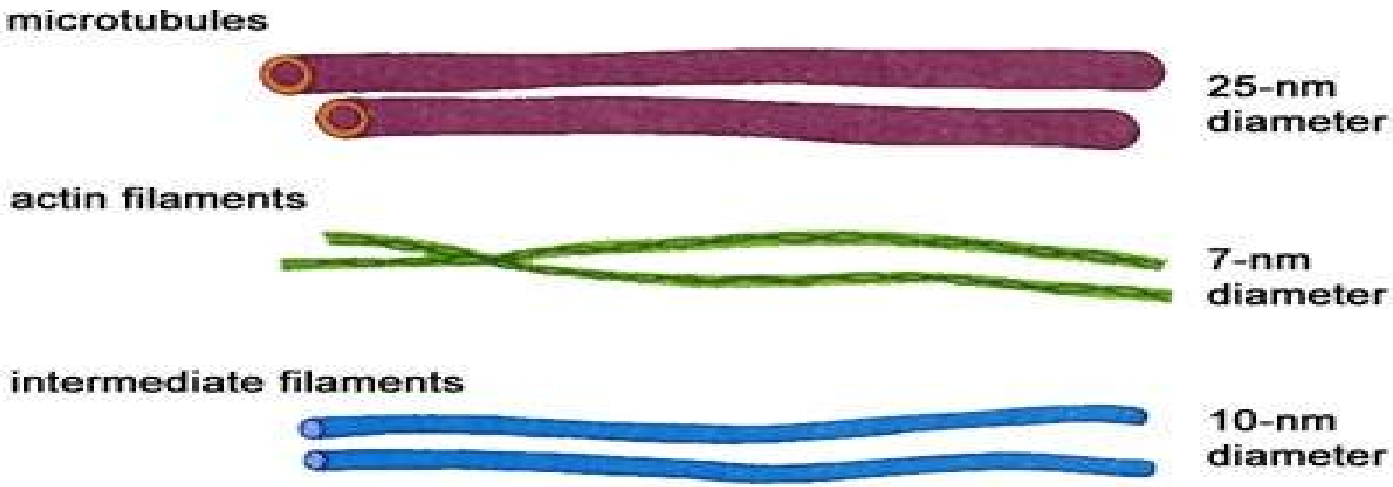


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CYTOSKELETON

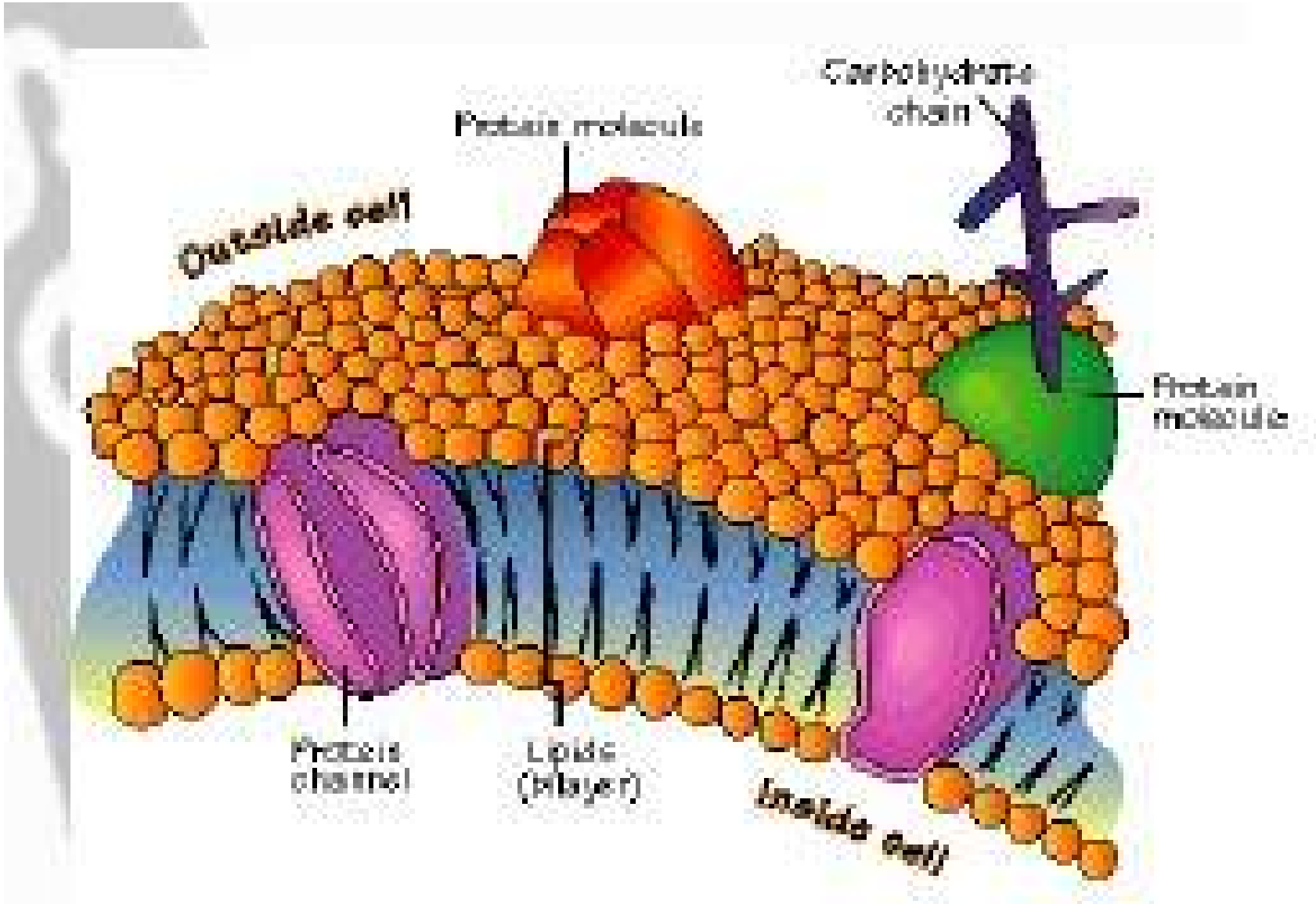
- The **cytoskeleton** is a series of intercellular proteins that help a cell with shape, support, and movement.
- It provides structural support to the cell
- These are group of fibrous, filamentous proteins and globular proteins
- **Cytoskeleton** has three main structural components: microfilaments, intermediate filaments, and microtubules

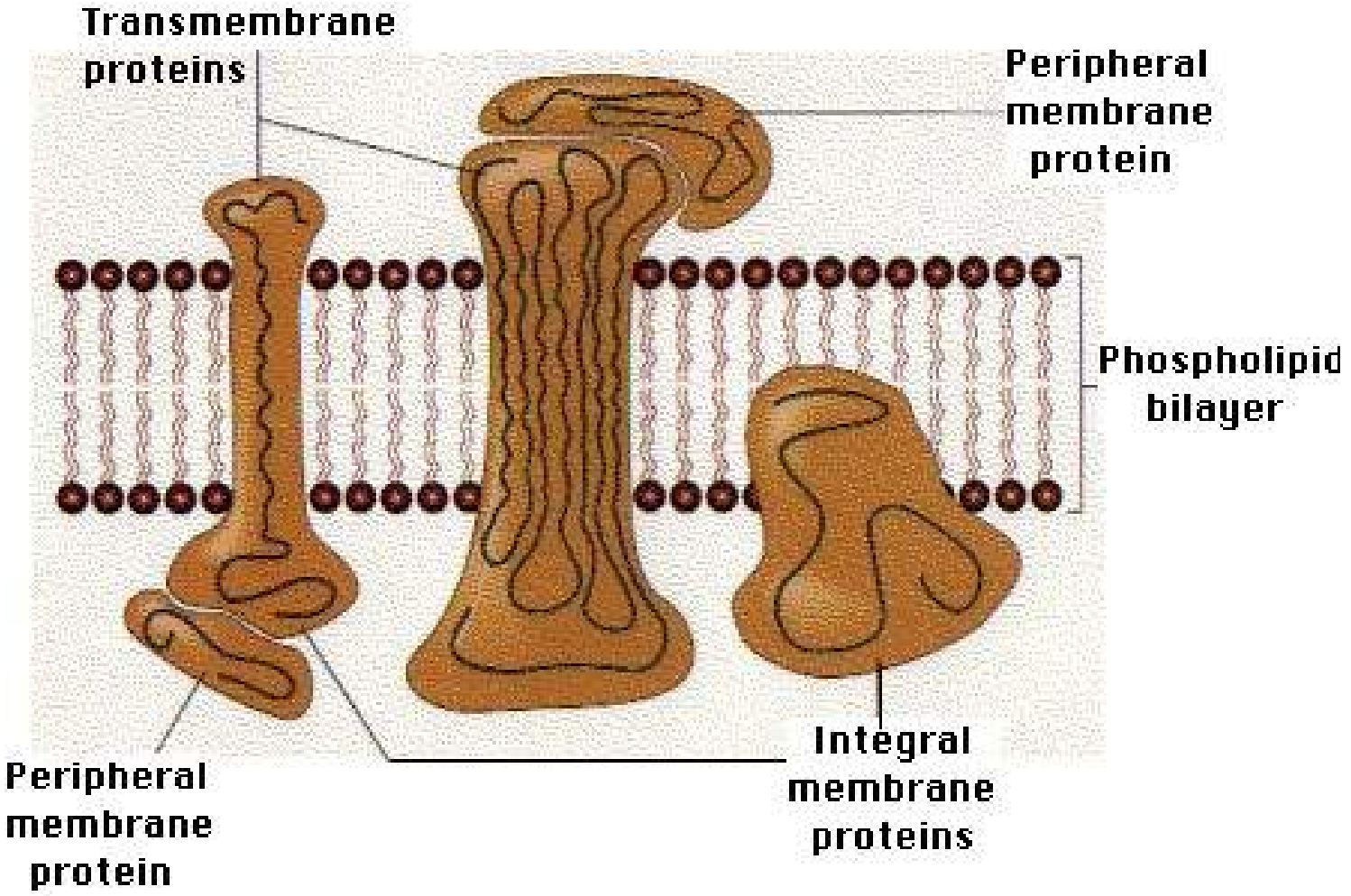

Plasma Membrane
Endoplasmic Reticulum
Ribosomes
Microfilaments and Intermediate Filaments


microtubules
25-nm diameter
actin filaments
7-nm diameter
intermediate filaments
10-nm diameter

Cell Membrane Proteins

- Transmembrane proteins
- Integral membrane proteins
- Peripheral membrane proteins


Carbohydrate chain
Protein molecule
Outside cell
Protein molecule
Protein channel
Lipids (bilayer)
Inside cell


Transmembrane proteins
Peripheral membrane protein
Phospholipid bilayer
Peripheral membrane protein
Integral membrane proteins

PROTEIN MISFOLDING

- Protein folding is a complex, trial-and-error process that can sometimes result in improperly folded molecules. leads to the formation of long, fibrillar protein assemblies consisting of β-pleated sheets as in **Amyloid disease**
- Accumulation of these insoluble, spontaneously aggregating proteins, called amyloids as in Alzheimer’s disease

Thank You